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Annals of Economics and Statistics, Number 144, December 2021

HOW COULD RUSSIA HAVE DEVELOPED WITHOUT THE REVOLUTION OF 1917?

IVAN KOROLEV[a]

This paper uses modern econometric approaches to program evaluation with one treated unit to predict how Russia could have developed without the Revolution of 1917. I construct the counterfactual for Russia for 1917–1938 using several variations of the synthetic control method, which is the preferred approach based on a number of placebo tests. My counterfactuals suggests that without the Revolution, Russia would probably have avoided the dramatic economic decline of the early 1920s and would have grown steadily throughout most of the 1920s. The predictions for the 1930s are less clear: while most specifications suggest that the counterfactual Russia would have reached a lower level of economic development than the Soviet Union, the results are sensitive to the choice of specification, and the placebo tests suggest that there is a lot of uncertainty about the results as the forecasting horizon becomes longer.

JEL Codes: C01, C53, N00, N14.
Keywords: Russian Revolution, Panel Data, Synthetic Control Method, Lasso.

1. INTRODUCTION

A major question in Russian economic history is how Russia could have developed in the 1920s and 1930s if the Russian Revolution had not happened in 1917. There are several studies that try to understand the consequences of various aspects of the Revolution for the economic development of Russia. In particular, Hunter and Szyrmer (2014) evaluate Stalin's economic policies in 1928–1940 using a multi-sector and multi-period linear model; Allen (2003) looks at the Soviet industrialization of the early 1930s and tries to understand what would have happened without centralized industrialization; Cheremukhin, Golosov, Guriev, and Tsyvinski (2017) investigate the effects of Stalin's industrialization on economic growth in Russia using a two-sector macroeconomic model.

All these studies primarily evaluate the effects of Stalin's economic policies in 1928–1940 rather than the Revolution itself. They also mostly rely on theory-based simulations, which have their advantages and disadvantages. On the one hand, this approach allows the authors to analyze particular policy changes and consider a wide range of scenarios; on the other hand, it might be difficult to model all factors that are relevant for economic growth (e.g. productivity, technology, international trade, and so on) together.

This paper uses modern econometric approaches to program evaluation with one treated unit to predict how Russia could have developed without the Revolution of 1917. More specifically, I use the approach proposed by Hsiao, Ching, and Wan (2012) based on a panel data regression, which I hereafter refer to as the HCW method; the approach based

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on the lasso regularization, developed by Tibshirani (1996), in a panel data regression; and the synthetic control method proposed by Abadie and Gardeazabal (2003), along with its modifications.

From an empirical point of view, I complement the existing literature on the Russian Revolution by studying its consequences using modern econometric techniques. I predict Russian GDP per capita after 1917 based on the pre-1917 data for Russia and both pre-1917 and post-1917 data for other countries. My counterfactuals can be viewed as an attempt to find a stable predictive relationship between Russia and other countries before 1917 and, assuming it would have remained unchanged after 1917, construct predictions according to this relationship.

For instance, one might think that because of international trade, technology spillovers, or other causes growth rates in different countries are interconnected. Then I could recover the relationship between the Russian economy and economies of other countries using econometric methods, without specifying the underlying structural model, and make forecasts based on this predictive relationship. In order to do so, I need to assume that the predictive relationship was stable over time and would not have changed after 1917 if the Revolution had not happened. I try to validate this assumption by conducting placebo tests, in which I construct the counterfactual GDP per capita series for the countries that did not experience revolutions and show that these counterfactual series are reasonably close to the actual series when my preferred methods are used.

My preferred counterfactuals suggest that without the Revolution, Russia would probably have avoided the dramatic economic decline of the late 1910s and early 1920s and would have grown steadily throughout the 1920s. The predictions for the 1930s are less clear. Most of my preferred specifications suggest that by the end of the 1930s, the counterfactual Russia would have reached a lower level of GDP per capita than the Soviet Union, but this result is fairly sensitive to the choice of specification. Moreover, the placebo tests suggest that the uncertainty about the counterfactuals increases as the forecasting horizon becomes longer, so any results about the 1930s should be viewed with caution.

From a methodological point of view, this paper contributes to the literature on the comparison of the HCW approach and the synthetic control method, such as Gardeazabal and Vega-Bayo (2017) and Wan, Xie, and Hsiao (2018). Compared with these papers, I consider a wider variety of approaches. In addition to the original versions of the HCW approach and the synthetic control method, I also utilize their various extensions and methods based on the lasso. I find that the synthetic control method generally performs better in the placebo tests than the HCW approach or the lasso-based methods.

This result is driven by finite sample, rather than asymptotic, considerations. Each of the methods I consider involves the bias-variance tradeoff: the more restrictions a given method imposes on the coefficients, the higher its bias but lower its variance. It appears that in the current setting, the variance dominates the bias, and successful methods are those that are able to control the variance and avoid overfitting.

The synthetic control method does precisely that by imposing more restrictions on the coefficients than other methods. In turn, because the number of potential controls is almost as large as the pre-treatment number of observations, both the HCW method and the lasso-based methods appear to suffer from overfitting: they generally fit the data very well in the pre-1917 period but produce unreasonable and overly variable out-of-sample forecasts. In fact, in some of the placebo tests, these methods are outperformed by a simple model

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based on a linear (in logarithms) time trend.

While I do not investigate the performance of various methods theoretically, these insights may carry over to other settings. Because the synthetic control method imposes the most stringent restrictions on the model coefficients, it is least likely to suffer from overfitting. Therefore, in settings with a large number of potential controls and a small number of pre-treatment observations, where overfitting is a primary concern, the synthetic control method may be preferred to other available methods.

The remainder of the paper is organized as follows. Section 2 gives a historical background of the Russian Revolution. Section 3 describes various approaches I use in this paper. Section 4 presents my results and discusses the placebo tests. Section 5 concludes. Appendix A contains additional results.

2. BACKGROUND

In this section I give a brief overview of the historical context of the Russian Revolution of 1917. In the second half of the 19th century and beginning of the 20th century Russia was a predominantly agrarian country, poorer than most European countries and approximately as rich as Japan. Russian industrial production was low in the middle of the 19th century (Gregory, 2014), but in the last quarter of the 19th century and the beginning of the 20th century Russia experienced industrial growth together with a rapid expansion of its rail system (Gregory (2014); Markevich and Nafziger (2017)). Nevertheless, as late as in 1913, around 75% of the labor force was employed in agriculture (Barber and Davies (1994)), and the agricultural sector comprised over 50% of GDP (Gregory (2004)). Overall, as noted by Davies (1994), the Tsarist Russian economy was a predominantly market one, though the state played a significant role in it.

Russia was facing economic and political challenges that, in particular, resulted in the Revolution of 1905. It led to a constitutional reform, but Russia remained a monarchy and Tsar Nicholas II stayed in power. The government, led by Prime Minister Pyotr Stolypin, started to implement large-scale agrarian reforms in 1906. However, Stolypin was assassinated in 1911, and the agrarian reforms were eventually cut short by World War I.

Russia entered World War I in 1914, and it exacerbated the existing economic, social, and political problems. For instance, Markevich and Harrison (2011) show that Russian GDP fell by 20% in 1917 relative to 1913. In 1917, the Revolution took place. Some historians, e.g. Davies (1994), believe that the Revolution actually consisted of two revolutions. The first happened in February–March 1917, when Tsar Nicholas II was forced to abdicate and the provisional government was formed. The second happened in October–November 1917, when the Bolsheviks came to power, replacing the provisional government. Others, e.g. Smith (2015) and Wade (2017), view the events of 1917–1918 as a single continuous revolution.

The Civil War (loosely speaking, the war between the Bolsheviks' Red Army and the Anti-Bolsheviks' White Army) started as a result of the Revolution. There is no universal agreement on the exact dates of the Civil War: historians believe it started in November 1917 or in the summer of 1918 and ended between 1920 and 1922 (see Bradley (1975); Mawdsley (2007); Bullock (2008)). The Bolsheviks won the war, and the Soviet Union was created in December 1922.

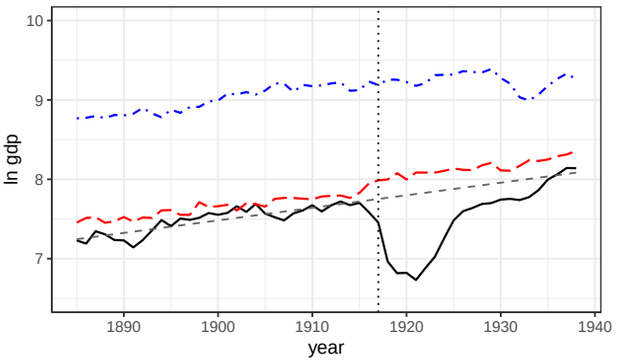
As the Bolsheviks came to power, they initially introduced the policy known as war communism in 1918. This policy involved wide-scale nationalization, rationing of consumer goods, compulsory acquisition of grain from peasants known as "prodrazverstka,"

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and many other restrictions on free markets (Richman (1981); Lih (1986); Davies (1994)). War communism, together with the Civil War, led to high inflation, widespread peasant disturbances, and a fuel and food crisis (Davies (1994)). As a result, in 1921 the Soviet government introduced the New Economic Policy (NEP) that reinstated substantial elements of the market economy. While the government continued to own large-scale industry, peasants were allowed to sell their products freely. Currency was soon stabilized, and after the disastrous famine of 1921–1922, the economy also recovered rapidly (Davies (1994)).

Starting in 1928, the government attempted centralized industrialization with a focus on capital goods industries together with collectivization in agriculture in order to quickly turn an agrarian country into an industrialized one. These policies were accompanied by the famine of 1932–1933, during which between 5.5 and 6.5 million people died (Wheatcroft and Davies (2004)). About 680 thousand people were then executed during the political repressions known as the Great Purge in 1936–1938 (Pipes (2001)).

Figure 1: Log GDP per Capita in Russia, Japan, and USA



Notes: Black solid line – Russian log GDP per capita. Light gray dashed line – the pre-1917 linear time trend in Russian log GDP per capita. Red dashed line – Japanese log GDP per capita. Blue dash-dotted line – the US log GDP per capita.

Figure 1 plots the evolution of Russian log GDP per capita over time in order to illustrate the main patterns of economic growth in Russia and the Soviet Union in 1885–1938 and compares it with the evolution of the US and Japanese log GDP per capita. I choose to plot the evolution of the US GDP per capita because the USA was one of the world's leading economies, so comparing the Russian economy to the US one helps illustrate the gap between Russia and the developed countries. In turn, Japan was pretty similar to Russia in terms of economic development before the Revolution, so the Japanese economy may serve as a useful benchmark for Russia after 1917. The figure also plots the pre-1917 linear time trend in Russian log GDP per capita as another benchmark for subsequent analysis. The linear time trend can serve as a very simple counterfactual to what might have happened after 1917 without the Revolution, but I try to improve upon it using modern econometric methods.

In this paper I estimate the counterfactual GDP per capita in Russia if the Russian Revolution had not taken place in 1917, and compare it with the actual GDP per capita. It is worth noting that I do not discuss whether it was possible for the Russian government to avoid the Revolution. It might be the case that in order to avoid the Revolution, it would have been necessary to implement political or economic reforms that would have changed the Russian growth path significantly.

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3. DATA AND METHODOLOGY

This section describes the data and methods that I use. My analysis is based on the updated 2020 version of the Maddison project, i.e. Bolt and van Zanden (2020), which contains data on Russian GDP per capita from 1885 on, as well as data on other countries. Table A1 in Appendix A lists all 31 countries for which complete GDP per capita series are available for 1885–1938; for other countries, the data for some or all years is missing. The data sources and the construction of the country characteristics are discussed in the table footnotes.

The series of Russian GDP for 1913–1928, as reported in Bolt and van Zanden (2020), was completed by Markevich and Harrison (2011). They built upon numerous previous studies, most notably Gregory (2004), to fill in the last gap in the 20th century Russian GDP series. An important concern about these data is that the territory and population of the country changed dramatically after the Revolution. One may also worry that the data quality for Russia and the Soviet Union might be affected by misreporting or the attempts of the government agencies to manipulate the data. However, Markevich and Harrison (2011) tried to address these issues when constructing their series. They explicitly took changes in territory and population into account in order to construct a consistent time series of GDP per capita for the Russian Empire and then the Soviet Union. Moreover, in their appendix,[1] Markevich and Harrison (2011) provided a detailed discussion of the quality of their estimates, the adjustments that they made to correct for possible misreporting or other data issues, and compared their estimates with those by other researchers.

In my analysis, I focus on the period of 1885–1938. 1885 is the earliest year when the data for Russia is available, while 1938 is the last year before the outbreak of World War II. Let $y_{j,t}$, $j = 1, ..., 31$, $t = 1, ..., 54$, be the outcome variable, such as the natural logarithm of GDP per capita, in country j in period t ($t = 1$ corresponds to 1885, while $t = 54$ corresponds to 1938). I order the countries so that $j = 1$ corresponds to Russia, the treated unit. The treatment takes place in 1917, i.e. at $t = T_0 + 1 = 33$. Thus, there are 32 time periods before treatment takes place and 22 periods after. The observed outcome is

$$y_{j,t} = d_{j,t} y^1_{j,t} + (1 - d_{j,t}) y^0_{j,t},$$

where $y^0_{j,t}$ is the potential outcome of unit j in year t if it is not treated and $y^1_{j,t}$ is its potential outcome if it is treated. $d_{j,t}$ is the treatment dummy equal to 1 for $j = 1$ and $t \geq 33$ and equal to 0 for all other observations. My goal is to predict how Russia could have developed without the Revolution, i.e. estimate $y^0_{1,t}$ or, alternatively, estimate the "treatment effect" $y^1_{1,t} - y^0_{1,t}$ for $t \geq 33$.

All methods that I use estimate the counterfactual $y^0_{1,t}$ using panel data by

$$\hat{y}^0_{1,t} = \hat{\alpha} + Y'_{-1,t}\hat{\beta}, \tag{3.1}$$

where $Y_{-1,t} = (y_{2,t}, ..., y_{J+1,t})'$, $J = 30$, are the outcomes for the countries in the control group, and $\hat{\alpha}$ and $\hat{\beta}$ are some estimates. These estimates are typically obtained from the estimated relationship between the observed outcome $y_{1,t}$ and the predictors $Y_{-1,t}$ using

[1] Available at `https://warwick.ac.uk/fac/soc/economics/staff/mharrison/data/greatwar/appendix.pdf`.

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the period before the Revolution.

However, estimating $\hat{\alpha}$ and $\hat{\beta}$ using the unrestricted regression may be problematic. From a statistical point of view, the primary concern is overfitting. Because the number of countries in the control pool, $J = 30$, is almost as large as the number of pre-treatment observations, $T_0 = 32$, estimating the unrestricted regression

$$y_{1,t} = \alpha + Y'_{-1,t}\beta + \varepsilon_{1,t}, \quad E[(1 \quad Y'_{-1,t})\, \varepsilon_{1,t}] = 0, \tag{3.2}$$

may result in unreliable estimates that have high variance. Thus, one needs to use some form of variable selection or regularization in order to avoid overfitting. The methods that I use differ in how exactly they impose restrictions or perform variable selection, and thus yield different estimates of $\hat{\alpha}$ and $\hat{\beta}$ and different counterfactuals.

From an economic point of view, using the parameters that solve the unrestricted problem may be unattractive because they may be difficult to interpret. Imposing certain constraints may make the resulting counterfactual easier to interpret. For instance, the synthetic control method requires that the parameters be non-negative and sum up to one, so that the synthetic unit can be viewed as a weighted average of the units in the control group.

Below I describe in detail different methods that I use.

3.1. *HCW Method*

The HCW approach (see, e.g. Hsiao, Ching, and Wan (2012) and Ke, Chen, Hong, and Hsiao (2017)) is based on selecting a subset of the countries in the control group and using them as regressors in equation (3.2), instead of using all available countries in the control group. One can use different variable selection methods. For instance, Hsiao, Ching, and Wan (2012) use best subset selection, while Shi and Huang (2019) suggest using forward selection.

Best subset selection considers all possible combinations of the predictors $Y_{-1,t}$ up to the size J_{max}. First, for any given number of predictors j, $1 \leq j \leq J_{max}$, the best model is selected based on the value of R^2. Then the resulting models, indexed $M^*(1), ..., M^*(J_{max})$, are compared based on a model selection criterion such as the Bayesian Information Criterion (BIC) or the corrected Akaike Information Criterion (AICc). The best model, i.e. the model with the lowest value of the BIC or AICc, is then used. In principle, the maximum number of regressors in a model, J_{max}, should not exceed the number of observations T_0 to avoid perfect multicollinearity. However, Vega-Bayo (2015) imposes a stricter restriction that J_{max} be no greater than $T_0 - 4$. It can also be set to a smaller number.

Alternatively, one could use forward stepwise selection. This algorithm starts from the null model, i.e. the model with the intercept term only, and adds the regressors to the model one by one. In each step, the most useful predictor is added to the model, and the variables that have been included in the model are never dropped. In other words, best subset selection considers all possible combinations of the predictors, while forward selection proceeds iteratively and only considers a subset of all possible models.

The main difference between the two approaches is computational (see Section 6.1 in James, Witten, Hastie, and Tibshirani (2013)). Because best subset selection involves estimating models with all possible combinations of the control variables, it is infeasible when the number of regressors is large. Forward selection solves this problem by pro-

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ceeding iteratively and fitting a smaller number of models. In my case, I can include up to $T_0 - 4 = 28$ variables in the model, so best subset selection remains feasible, and I use best subset selection. I try using both the Bayesian Information Criterion (BIC) and the corrected Akaike Information Criterion (AICc) to select the optimal number of variables. After the appropriate regressors are selected, the resulting model is estimated using ordinary least squares (OLS).

I implement this method using the **pampe** package in R by Vega-Bayo (2015).

3.2. *Lasso Method*

Alternatively, one can use the lasso instead of variable selection. It was first developed by Tibshirani (1996), and Li and Bell (2017) suggest using it in the context of program evaluation. The lasso shrinks the coefficients to zero and sets some of them exactly to zero, so essentially it automatically performs variable selection. One can also use the post-lasso OLS: first, select the appropriate control variables using the lasso, then run the OLS regression with the selected regressors only. The lasso solves the following problem:

$$\min_{\alpha,\beta} \sum_{t=1}^{T_0} (y_{1,t} - \alpha - Y'_{-1,t}\beta)^2 + \lambda \sum_{j=2}^{J+1} |\beta_j|,$$

where λ is a tuning parameter. The higher the value of λ, the more the resulting coefficients will be shrunken to zero. The value of λ can be selected using leave-one-out cross-validation, as in Li and Bell (2017). For each value λ, I compute the averaged squared error over all pre-treatment observations:

$$CV(\lambda) = \frac{1}{T_0} \sum_{t=1}^{T_0} e_t(\lambda)^2 = \frac{1}{T_0} \sum_{t=1}^{T_0} (y_{1,t} - \hat{\alpha}_{-t}(\lambda) - Y'_{-1,t}\hat{\beta}_{-t}(\lambda))^2,$$

where $\hat{\alpha}_{-t}(\lambda)$ and $\hat{\beta}_{-t}(\lambda)$ are the leave-one-out estimates of α and β that solve the lasso or post-lasso problem without using the observation t. Then I choose λ^* that minimizes $CV(\lambda)$.

I implement the lasso and post-lasso OLS using the **LassoOLS** command in the **HDCI** package in R by Liu, Xu, and Li (2020) (see also Liu and Yu (2013)). I also utilize some alternative methods for tuning parameter selection that I discuss later.

3.3. *Synthetic Control Method*

The synthetic control method is based on calculating a convex combination of the units in the donor pool that best replicates the treated unit in the pre-treatment period. The counterfactual is then based on this synthetic unit. The weights in the synthetic unit are required to sum up to one. As a result, it can be viewed as a weighted average of the control units. This requirement may seem somewhat ad hoc, and Doudchenko and Imbens (2016) discuss various possible alternative restrictions on the weights. However, the restrictions that the synthetic control method imposes make the results easier to interpret, which might explain why it has become popular in applied work (see, e.g., Abadie and Gardeazabal (2003), Abadie, Diamond, and Hainmueller (2010), Pinotti (2015)).

In my description of the synthetic control method, I follow the exposition of Doudchenko and Imbens (2016), because their setting is closest to mine. In particular, I do

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not have any additional regressors based on which I could match the treated and control units. I also assign the same weight to all years before 1917 (in other words, use the ordinary least squares objective function rather than the weighted least squares objective function) because there is no ex ante reason to assign different weights to different years. The synthetic control weights β_j solve:

$$\min_{\beta} \sum_{t=1}^{T_0} (y_{1,t} - Y'_{-1,t}\beta)^2 \quad \text{subject to } \sum_{j=2}^{J+1} \beta_j = 1, \beta_j \geq 0, j = 2, ..., J+1$$

The counterfactual is given by

$$\hat{y}^0_{1,t} = Y'_{-1,t}\hat{\beta}_{sc},$$

where $\hat{\beta}_{sc}$ are the synthetic control coefficients (weights).

One potential drawback of the synthetic control method is that it rules out a parallel shift in trends (see, e.g., Doudchenko and Imbens (2016)). In other words, if the treated country and all countries in the donor pool have parallel trends but different levels of the outcome variable, then the synthetic control method will fail to find a good match. In order to solve this issue, one could include the intercept term in the synthetic control problem. Alternatively, one could demean the outcome variable in the pre-treatment period for each unit, i.e. compute $\tilde{y}_{j,t} = y_{j,t} - \bar{y}_j$, where $\bar{y}_j = T_0^{-1} \sum_{t=1}^{T_0} y_{j,t}$, and then apply the synthetic control method to the demeaned data:

$$\min_{\beta} \sum_{t=1}^{T_0} (\tilde{y}_{1,t} - \tilde{Y}'_{-1,t}\beta)^2 \quad \text{subject to } \beta_j \geq 0, \sum_{j=2}^{J+1} \beta_j = 1,$$

where $\tilde{Y}_{-1,t} = (\tilde{y}_{2,t}, ..., \tilde{y}_{J+1,t})'$. The counterfactual becomes

$$\hat{y}^0_{1,t} = \bar{y}_1 + \tilde{Y}'_{-1,t}\tilde{\beta}_{sc},$$

where $\tilde{\beta}_{sc}$ are the synthetic control coefficients obtained using the demeaned data.

I implement the synthetic control method using the **Synth** package in R by Abadie, Diamond, and Hainmueller (2011).

4. RESULTS

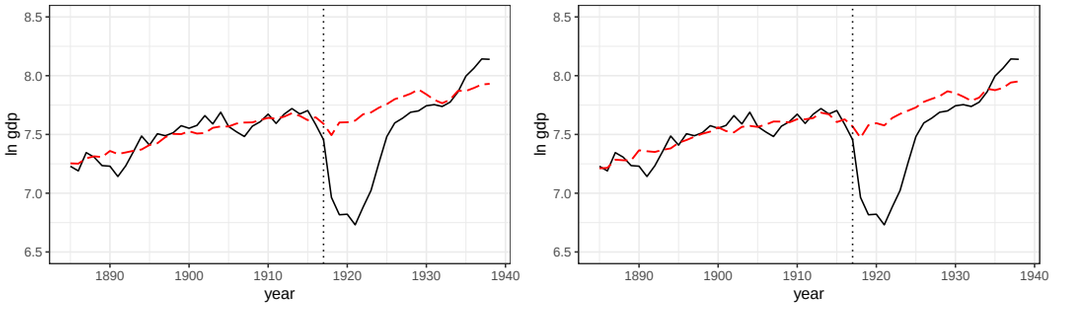
In this section, I present my empirical results. First, I discuss the counterfactual from the synthetic control method, as it is preferred based on my placebo tests. I then turn to the robustness checks based on different versions of the synthetic control method and show that the counterfactual growth path of Russia in the 1920s is fairly robust to the assumptions made, while the results for 1930s are more sensitive. Next, I turn to the placebo tests that use backdating in Russia or 1917 in other countries as hypothetical interventions and analyze the performance of different methods. I then present the results from the HCW method and from the lasso-based methods. Additional results are presented in Appendix A.

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4.1. *Main Results*

The left panel of Figure 2 displays the actual Russian log GDP per capita and the counterfactual from the standard version of the synthetic control method. The right panel presents the counterfactual from the demeaned version of the synthetic control method. Both versions of the synthetic control method predict a decline in GDP per capita during and immediately after World War I, economic growth during the 1920s, then a recession in the beginning of the 1930s, possibly related to the Great Depression, and finally recovery. According to this prediction, the counterfactual unit outperforms the actual one during the 1920s and early 1930s, but the actual Soviet Union overtakes the counterfactual unit in the second half of the 1930s.

Figure 2: Actual and Counterfactual Log GDP per Capita, Synthetic Control

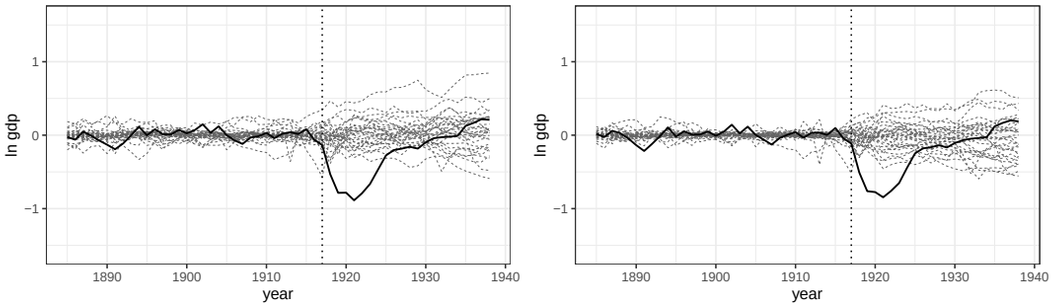


Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. Left: synthetic control method. Right: synthetic control method, demeaned.

While I do not formally conduct inference on my estimates, I follow the approach of Abadie, Diamond, and Hainmueller (2010) and conduct a sequence of placebo tests to assess the significance of my estimates and place them into context. More specifically, I apply the synthetic control method to each of the 30 countries in the donor pool, as if the revolution would have happened in one of the other countries in the dataset, rather than in Russia. Then I construct the counterfactual unit and compute the gap between the actual and counterfactual log GDP per capita, $y_{j,t} - \hat{y}_{j,t}$. Figure 3 presents the results graphically. The log GDP per capita gap for Russia is plotted as a solid line, while the gaps for the placebo units are plotted as lighter dashed lines. When applying the synthetic control method to the placebo units, I drop Russia from the donor pool. I repeat this exercise, while keeping Russia in the donor pool, and present the results in Figure A1 in Appendix A. They remain virtually unchanged.

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Figure 3: Log GDP Gaps, Russia and Placebo Units, Synthetic Control



Notes: Black solid line – difference between the actual and counterfactual log GDP per capita for Russia. Gray dashed lines – differences between the actual and counterfactual log GDP per capita for 30 placebo units. Left: synthetic control method. Right: synthetic control method, demeaned.

As we can see from the figure, both the standard version of the synthetic control method and especially its demeaned version are able to fit the actual pre-1917 log GDP per capita paths fairly well for most countries. After 1917, the gaps gradually diverge over time, implying that the uncertainty about the estimates increases as one attempts to extrapolate farther into the future. As for the Russian log GDP per capita gap, it is negative with a large absolute value and lies at the very bottom of the placebo gaps distribution after 1917 and throughout most of the 1920s, suggesting that Russia might have developed faster without the Revolution than it actually did. As for the 1930s, the gap becomes close to zero and then positive, suggesting that the Soviet Union might have outperformed the counterfactual Russia. However, the log GDP gap for Russia lies close to the middle of the placebo gap distribution, which itself is fairly dispersed. Thus, the results for the 1930s contain more uncertainty and should be viewed with more caution than the results for the 1920s.

TABLE I

SYNTHETIC CONTROL, BALANCING

Unit	Population (thousands)	WWI Losses (thousands)	WWI Losses (synthetic, %)	WWI Losses (% of population)
Actual	156,192	1,700	1.09	1.09
Synthetic	65,922	20.5	0.30	0.03
Synthetic, DM	3,813	8.7	0.28	0.23
Synthetic, no Finland	74,172	13.0	0.01	0.02
Synthetic, no Finland, DM	4,074	0.3	0.005	0.007
Synthetic, WWI, DM	31,481	735.9	1.62	2.34

Notes: The table compares the characteristics of Russia with the characteristics of the synthetic control units in various specifications (baseline; demeaned; Finland is dropped from the donor pool; demeaned and Finland is dropped from the donor pool; demeaned with only the countries that lost at least 0.5% of their population in World War I in the donor pool). "DM" stands for "demeaned." WWI Losses (synthetic, %) are computed as the weighted average of percentage World War I losses of the countries in the synthetic control unit. WWI Losses (% of population) are computed as the ratio of the synthetic World War I losses to the synthetic population.

Table I shows the characteristics of Russia and the synthetic control unit. The population of the synthetic control unit is much smaller than the population of actual Russia. This is not surprising, given that Russia was one of the most populous countries in the world

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at the time, and India is the only country in my dataset with a larger population than Russia. Thus, I need to assume that all economies were "scalable" (or, in economic terms, that there were constant returns to scale), so that Russia could be represented as a small country "replicated" many times, and that the population size, or population density, does not matter for economic growth.

TABLE II

SYNTHETIC CONTROL WEIGHTS

	Baseline		No Finland		WWI	
Country	Synthetic	Synthetic (demeaned)	Synthetic	Synthetic (demeaned)	Synthetic	Synthetic (demeaned)
Argentina	0.052	-	0.144	-	-	-
Canada	-	0.004	-	-	-	-
Denmark	-	0.249	-	0.319	-	-
Finland	0.332	0.316	-	-	1	0.486
Germany	-	-	-	-	-	0.338
India	0.206	-	0.231	-	-	-
Norway	0.001	-	-	-	-	-
Peru	0.115	-	0.152	-	-	-
Portugal	-	0.009	-	0.038	-	-
Sri Lanka	0.289	0.410	0.373	0.434	-	-
Sweden	0.003	-	0.098	-	-	-
Switzerland	-	0.012	-	0.209	-	-
UK	-	-	-	-	-	0.175

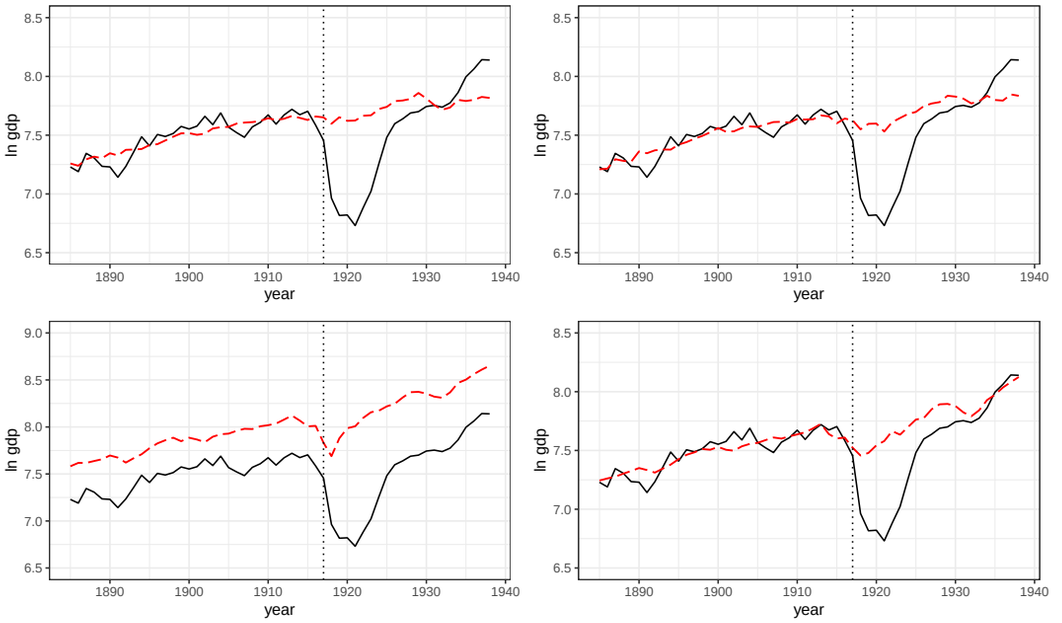
Notes: The table presents the composition of the synthetic control units in various specifications. "Baseline": includes all available countries in the donor pool. "No Finland": drops Finland from the donor pool. "WWI": only includes the countries that lost at least 0.5% of their population in World War I (Austria, Finland, France, Germany, Italy, the United Kingdom, Australia, New Zealand, and Canada) in the donor pool.

Table II reports the weights that the synthetic control unit assigns to different countries. As we can see, the synthetic unit assigns high weight to Finland, as well as to some non-European countries. Both these features might be problematic.

As for Finland, it was actually a part of the Russian Empire before the Revolution. While it is reasonable to think that it was affected by the Russian Revolution (for instance, Finland became independent in 1917), I keep it in the pool of potential controls in my baseline analysis. Communists did not come to power in Finland, so Finland may serve as a reasonable counterfactual to what might have happened in Russia if it had remained a market economy. To study whether my results are sensitive to the inclusion of Finland, I drop it from the donor pool and repeat the analysis. The upper panel of Figure 4 presents the results. While the composition of the synthetic control unit changes, the growth path of the synthetic unit remains fairly similar to that in the baseline analysis.

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Figure 4: Actual and Counterfactual Log GDP per Capita, Robustness



Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. Top left: synthetic control method, Finland is dropped from the donor pool. Top right: synthetic control method, demeaned, Finland is dropped from the donor pool. Bottom left: synthetic control method, only the countries that lost at least 0.5% of their population in World War I in the donor pool. Bottom right: synthetic control method, demeaned, only the countries that lost at least 0.5% of their population in World War I in the donor pool.

As for non-European countries, one may be worried that these countries, which include Argentina, India, Peru, and Sri Lanka, might have been very different from Russia. It might have been useful to have Eastern European countries, such as Poland, Bulgaria, Czechoslovakia, or Romania, in the donor pool, because those countries might have been more similar to Russia and might have served as better controls. Unfortunately, the data on GDP per capita for these countries is not available in the Maddison project. This may be due to the fact that many of these countries were not independent at the time and were parts of Russia, Germany, or Austria-Hungary. As a result, I cannot include these countries in the donor pool.

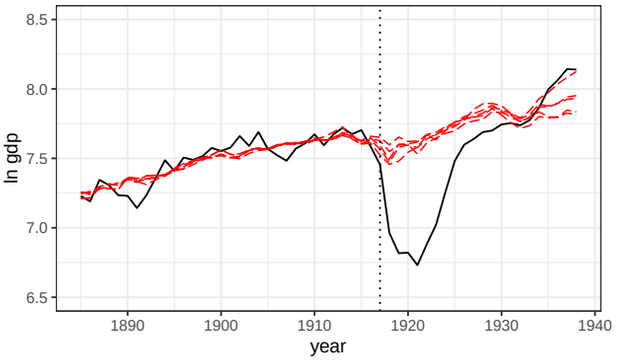
Because the synthetic control unit assigns high weight to non-European countries, it may also fail to properly account for the effect of World War I on the Russian economy. Indeed, Finland and Canada are the only countries selected by the baseline synthetic control method in Table II that suffered heavy war casualties, around 0.8–0.9% of their population (see Table A1). As shown in Table I, the synthetic control unit would lose between 0.03% and 0.3% of its population, depending on whether the usual or demeaned version is used and on whether the counterfactual losses are computed as the weighted average of the actual percentage losses of each county involved or as the ratio of the counterfactual losses and population. In contrast, Russia lost 1.09% of its population due to the war. Thus, the resulting synthetic control unit may underestimate the negative effects of World War I on the economy.

To deal with this concern, I restrict the donor pool to the countries that lost at least 0.5% of their population in World War I. This pool of countries consists of Austria, Finland, France, Germany, Italy, the United Kingdom, Australia, New Zealand, and Canada.

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I repeat my analysis using the synthetic control method and its demeaned version with this restricted donor pool. The bottom panel of Figure 4 presents the results. The standard version of the synthetic control method is unable to replicate Russian log GDP per capita before the Revolution, because all countries in the donor pool were significantly richer than Russia. But the demeaned version of the synthetic control method, which allows for the parallel shift of trends, is able to replicate the Russian economy before the Revolution well.

Figure 5: Range of Counterfactual Scenarios



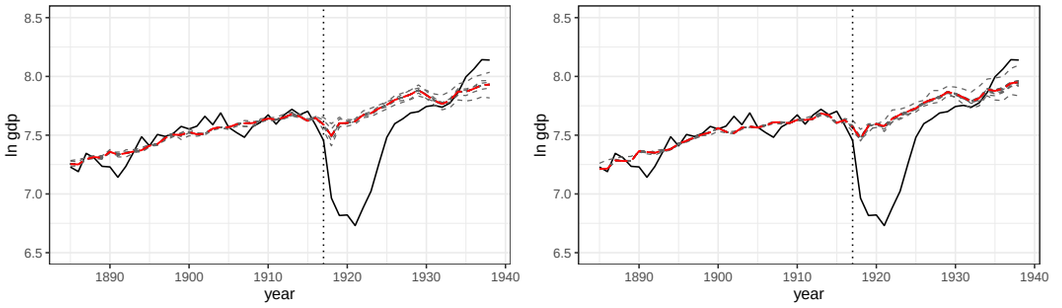
Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita from the synthetic control method and its four modifications: the demeaned version, the "no Finland" version, the "demeaned no Finland" version, and the "demeaned WWI" version.

Figure 5 plots the counterfactual growth paths from the synthetic control method and its four modifications: the demeaned version, the version that drops Finland from the donor pool, the demeaned version that drops Finland from the donor pool, and the demeaned version that restricts the donor pool to the countries that suffered heavy World War I losses. These five counterfactuals correspond to the ones in Figure 2 and in the two upper panels and the lower right panel of Figure 4. As we can see, these five counterfactuals provide an almost identical fit of Russian GDP per capita before the Revolution. After the Revolution, they differ somewhat in the extent of the economic downturn that would have happened in the end of the 1910s as a result of World War I, but they all yield fairly similar growth paths for most of the 1920s and early 1930s. There is more disagreement between the different counterfactuals in the end of the 1930s: some of them predict that the counterfactual unit would have reached the same GDP per capita level as the actual Soviet Union, while others are much more pessimistic.

These results appear to be in line with the findings presented in Figure 3. Given that the uncertainty about the estimates increases as the forecasting horizon becomes longer, it is not surprising that one can be much less certain about what could have happened in Russia without the Revolution in the late 1930s than in the 1920s.

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Figure 6: Actual and Counterfactual Log GDP per Capita, Leave-one-out Estimates



Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita from the synthetic control method. Gray dashed lines – counterfactual log GDP per capita from the synthetic control method that drops one of the countries from the donor pool. Left: synthetic control method. Right: synthetic control method, demeaned.

As a final robustness check, I perform the leave-one-out synthetic control method estimation as suggested in Abadie (2019). I remove each country from the donor pool one by one and apply the synthetic control method and its demeaned version with the modified donor pool. This exercise examines whether my results are sensitive to the inclusion of any particular country in the donor pool. As we can see from Figure 6, the results seem to be fairly robust to the composition of the donor pool.

4.2. *Placebo Tests*

Above, I have used the placebo tests involving other countries to quantify the uncertainty about the synthetic control estimates. In this section, I take a closer look at the placebo tests. First, I use backdating in Russia to study the sensitivity of my results to the choice of the matching period. Then I use backdating in Russia as well as 1917 in other countries as hypothetical interventions to compare the performance of different methods.

4.2.1. *Backdating*

Backdating, as discussed in Abadie (2019), refers to using Russia itself as a placebo, but changing the date of the intervention as if the Revolution would have happened earlier. I consider three such dates: 1905, 1909, and 1913. For instance, when I backdate the intervention to 1909, I use the period of 1885–1908 to estimate the synthetic control weights and then construct the counterfactual for the period of 1909–1916 and beyond. This exercise has two goals: first, to study how robust the counterfactual for the period after the Revolution is to the choice of the matching period; second, to check how well the counterfactual matches the actual log GDP per capita for the period after the end of the matching period but before the Revolution.

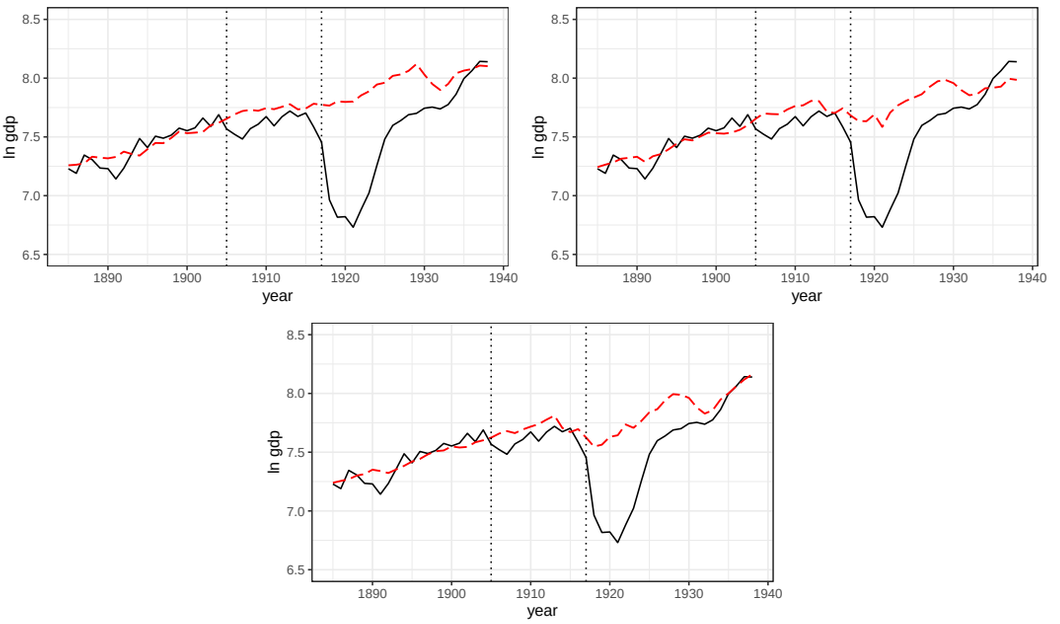
I should note that it is questionable whether 1905 can serve as a reasonable placebo test. As discussed above, a revolution actually did take place in Russia in 1905, and the government started implementing large-scale reforms as a result. Thus, it would be unreasonable to think that no intervention took place in 1905. If anything, the difference between the counterfactual and actual unit in the period of 1905–1916 might illustrate the consequences of the Revolution of 1905 for the Russian economy.

Figures 7, 8, and 9 present the results from the synthetic control method when the intervention is backdated to 1905, 1909, and 1913, respectively. First, as we can see,

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when the intervention is backdated to 1905, the synthetic control method fails to match the path of Russian GDP per capita in 1905–1916. It appears that economic growth in Russia slowed down after the Revolution of 1905, as compared to the growth forecasted based on the pre-1905 period. Therefore, it appears that both the Revolution of 1905 and the Revolution 1917 were associated with economic slowdowns, though I should stress again that these revolutions were endogenous and one should be very careful with causal interpretation of these results.

Figure 7: 1905 as Placebo

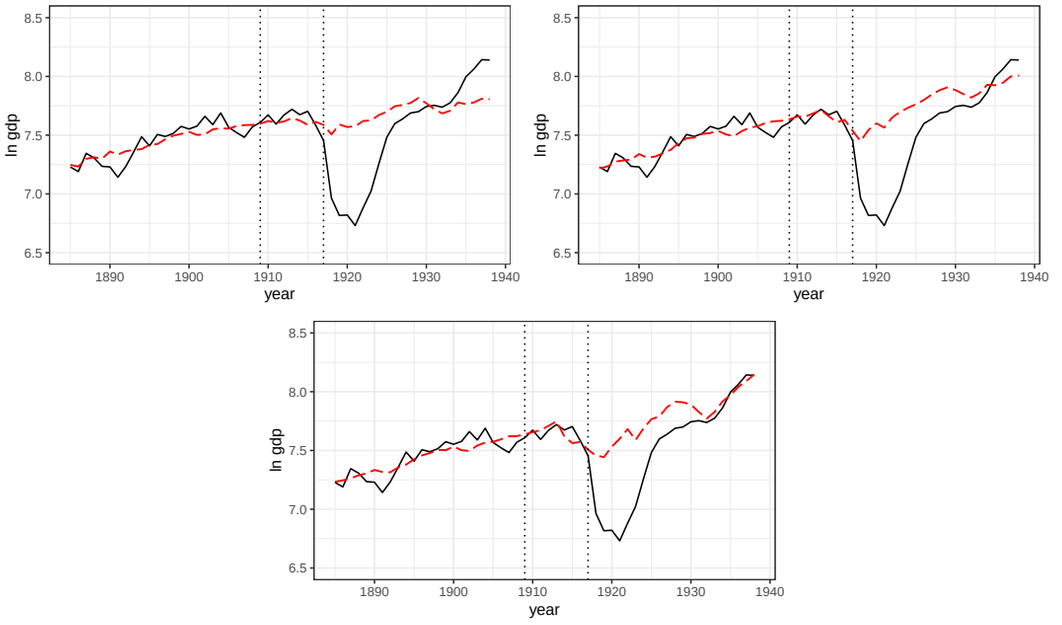


Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. Top left: synthetic control method. Top right: synthetic control method, demeaned. Bottom: synthetic control method, only the countries that lost at least 0.5% of their population in World War I in the donor pool.

When the intervention is backdated to 1909 or 1913, the standard version of the synthetic control method underpredicts Russian GDP per capita after the backdated intervention and before the actual Revolution. It also fails to predict the decline in Russian GDP per capita around 1915–1917 that presumably happened because of World War I. The demeaned version of the synthetic control method that only includes the countries that suffered heavy World War I losses in the donor pool does predict this decline, but it starts too early: around 1912 or 1913 rather than 1915.

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Figure 8: 1909 as Placebo



Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. Top left: synthetic control method. Top right: synthetic control method, demeaned. Bottom: synthetic control method, only the countries that lost at least 0.5% of their population in World War I in the donor pool.

In turn, the demeaned version of the synthetic control method that includes all countries in the donor pool seems to adequately match the path of Russian GDP per capita prior to the Revolution of 1917, both when the intervention is backdated to 1909 and to 1913. It also predicts a fairly substantial economic decline right around 1915–1917. On a final note, different methods with different backdating periods seem to yield more or less similar predictions about the 1920s, while there is much more uncertainty about the 1930s, in line with the results discussed above.

4.2.2. *Ranking of Different Methods*

Placebo tests that utilize backdating can also be used to compare different methods. So far, I have focused on the synthetic control method, and in this section I demonstrate why I prefer it to other methods. I backdate the intervention for Russia to 1905, 1909, or 1913, apply different methods to obtain the counterfactual for the period up to 1916, and then compute the mean squared error as a measure of fit. Recall that T_0 corresponds to 1916, and suppose that T_{-1} corresponds to the backdated intervention. Then the mean squared error for the placebo test is given by

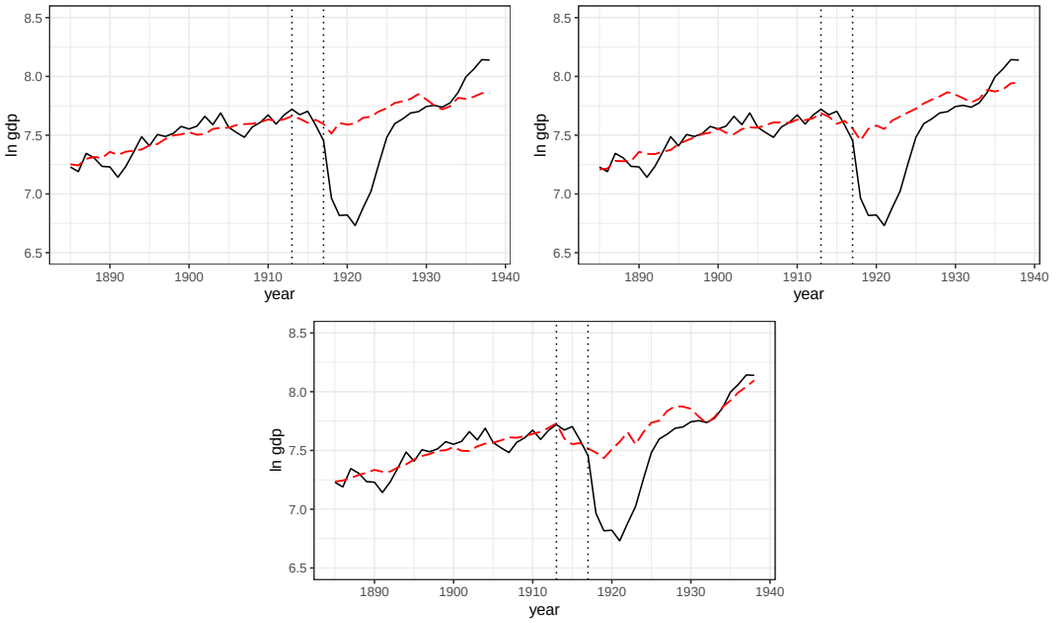
$$\frac{1}{T_0 - T_{-1}} \sum_{t=(T_{-1}+1)}^{T_0} (y_{1,t} - \hat{y}_{1,t})^2,$$

where $y_{1,t}$ is the actual Russian log GDP per capita in period t and $\hat{y}_{1,t}$ is its estimate from a particular method.

Table III presents the values of the placebo mean squared error, multiplied by 1,000, from different methods. It uses different donor pools (all countries or only those that suffered heavy World War I losses) and considers a number of different methods. In addition

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Figure 9: 1913 as Placebo



Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. Top left: synthetic control method. Top right: synthetic control method, demeaned. Bottom: synthetic control method, only the countries that lost at least 0.5% of their population in World War I in the donor pool.

to the synthetic control method and its demeaned version, it considers four varieties of the HCW method: one that allows for up to $J_{max} = 8$ predictors and uses the AICc to select the optimal number of variables; one with $J_{max} = 8$ and the BIC; one with $J_{max} = 28$ and the AICc; and one with $J_{max} = 28$ and the BIC.

Next, I consider the lasso and post-lasso OLS with different choices of the tuning parameter. As a benchmark, I use leave-one-out cross-validation (LOOCV), as in Li and Bell (2017). However, Burman, Chow, and Nolan (1994) and Racine (2000) suggest alternative types of cross-validation that can be used with dependent data. These methods typically involve removing blocks of data, rather than individual observations, in order to reduce dependence. Thus, I try splitting my data into 4 blocks of 8 subsequent observations each and performing 4-fold cross-validation using the resulting blocks, or splitting the data into 8 blocks of 4 subsequent observations each and performing 8-fold cross-validation using the resulting blocks.

In addition to the methods above, I also include the counterfactual based on a linear time trend as a benchmark. This counterfactual involves estimating the linear (in logarithms) time trend in the pre-intervention period and then extending it into the future.

As we can see from Table III, for all three placebo years it is one of the varieties of the synthetic control method that performs best: for 1909 and 1913, it is the demeaned synthetic control method; for 1905, it is the demeaned synthetic control method with the restricted donor pool.

The HCW method with $J_{max} = 28$ seems to be the worst one. The ranking of the HCW method with $J_{max} = 8$ and the lasso or post-lasso OLS depends on the backdating year. For the lasso and post-lasso OLS, there is no clear ranking of different choices of the tuning parameter. Overall, based on the placebo tests for Russia, the synthetic control method appears to be the preferred approach.

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TABLE III

RUSSIA AS PLACEBO: $1,000 \times MSE$

	Placebo Year: 1905 Donor Pool:		Placebo Year: 1909 Donor Pool:		Placebo Year: 1913 Donor Pool:	
Method	All	WWI	All	WWI	All	WWI
Synthetic	17.568	163.704	3.509	153.698	4.083	147.349
Synthetic, demeaned	17.096	**11.105**	**2.013**	4.071	**3.542**	7.099
HCW, AICc, $J_{max} = 8$	19.615	110.217	16.522	16.070	75.278	24.168
HCW, BIC, $J_{max} = 8$	19.615	110.217	16.522	16.070	75.278	24.168
HCW, AICc, $J_{max} = 28$	9772.746	-	16.522	-	157.040	-
HCW, BIC, $J_{max} = 28$	7922.848	-	1015.163	-	936.902	-
Lasso, LOOCV	48.754	91.424	38.507	6.473	28.493	14.836
Post-Lasso, LOOCV	39.738	68.363	20.312	12.171	51.932	20.015
Lasso, 4-fold CV	48.754	26.428	46.099	12.708	6.689	8.665
Post-Lasso, 4-fold CV	39.738	93.127	50.860	8.513	5.038	5.923
Lasso, 8-fold CV	-	-	38.507	11.096	-	-
Post-Lasso, 8-fold CV	-	-	104.379	8.513	-	-
Trend	52.006	-	11.565	-	9.918	-

Notes: The table reports the mean squared error (MSE), multiplied by 1,000, for different placebo years for Russia: 1905, 1909, and 1913. For a given placebo year T_{-1}, the each method is applied to the data prior to the placebo year, the counterfactuals are obtained for the period from the placebo year to 1916, and then the MSE is computed as $\frac{1}{T_0 - T_{-1}} \sum_{t=(T_{-1}+1)}^{T_0} (y_{1,t} - \hat{y}_{1,t})^2$, where T_0 corresponds to 1916. See the text for the detailed description of different methods. Columns marked "WWI" only include the countries that lost at least 0.5% of their population in World War I (Austria, Finland, France, Germany, Italy, the United Kingdom, Australia, New Zealand, and Canada) in the donor pool.

Next, I turn to the placebo tests based on other countries. I drop Russia from my data, then one by one treat each country in the donor pool as a treated unit and the remaining countries as control units, estimate the model using the period of 1885–1916, and obtain the counterfactuals for 1917–1938. Then I compute the placebo MSE for each country j as $\frac{1}{22} \sum_{t=33}^{54} (y_{j,t} - \hat{y}_{j,t})^2$ and average the resulting MSEs over all countries in the donor pool.

For the lasso and post-lasso OLS, in addition to LOOCV and k-fold cross-validation discussed above, I also leverage the availability of the placebo tests and use them to select the tuning parameter. I call this procedure "placebo-based cross-validation." I first pick a grid of possible values of the lasso tuning parameter λ. Next, for each possible value of λ, I estimate the model and compute the corresponding placebo MSE for each country in the donor pool. Then I average the country-specific placebo MSEs over all countries and pick the value of λ that minimizes the resulting average placebo MSE. As shown in Table IV, this approach results in a significantly higher value of λ than leave-one-out cross-validation. Higher values of λ mean more regularization, which reduces the amount of in-sample overfitting and reduces the variance of out-of-sample forecasts.

The values of the placebo mean squared error, multiplied by 1,000, are presented in Table V. As we can see, the synthetic control method and its demeaned version once again demonstrate the best performance in this test, while all methods utilizing the HCW approach perform much worse than the simple benchmark based on the linear (in logarithms) time trend. The lasso-based approaches with LOOCV and k-fold CV are generally comparable to the approach based on the linear time trend. The lasso-based approaches with placebo-based cross-validation perform somewhat better. One should be cautious

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TABLE IV

VALUES OF λ SELECTED BY CROSS-VALIDATION

	Lasso	Post-Lasso OLS
Leave-one-out CV	0.0076	0.0175
4-fold CV	0.0048	0.0083
8-fold CV	0.0040	0.0100
Placebo-based CV	0.0298	0.0423

Notes: The table presents the values of the lasso shrinkage parameter λ selected by different types of cross-validation. See the text for the detailed description of different types of cross-validation.

with interpreting this result as clear evidence in favor of placebo-based cross-validation, because it directly minimizes the average placebo MSE. Hence, it is not surprising that it outperforms other types of cross-validation in terms of the average placebo MSE. Overall, the placebo tests based on other countries suggest that the synthetic control method, and especially its demeaned version, is the preferred approach.

TABLE V

OTHER COUNTRIES AS PLACEBO: $1,000 \times MSE$

Method	Donor Pool: All
Synthetic	43.943
Synthetic, demeaned	**43.697**
HCW, AICc, $J_{max} = 8$	146.442
HCW, BIC, $J_{max} = 8$	159.959
HCW, AICc, $J_{max} = 28$	143.659
HCW, BIC, $J_{max} = 28$	254.004
Lasso, LOOCV	67.097
Post-Lasso, LOOCV	72.150
Lasso, 4-fold CV	61.157
Post-Lasso, 4-fold CV	64.877
Lasso, 8-fold CV	73.588
Post-Lasso, 8-fold CV	61.755
Lasso, placebo-based CV	53.568
Post-Lasso, placebo-based CV	57.387
Trend	65.015

Notes: The table reports the mean squared error (MSE), multiplied by 1,000, when the countries in the donor pool are used as placebos. Each method is applied to the data prior to 1917, then counterfactuals are obtained for the period from 1917 to 1938, and the placebo MSE for each country j is given by $\frac{1}{22}\sum_{t=33}^{54}(y_{j,t} - \hat{y}_{j,t})^2$, the reported MSEs are averaged over all countries in the donor pool. See the text for the detailed description of different methods.

4.3. *Results from Different Methods*

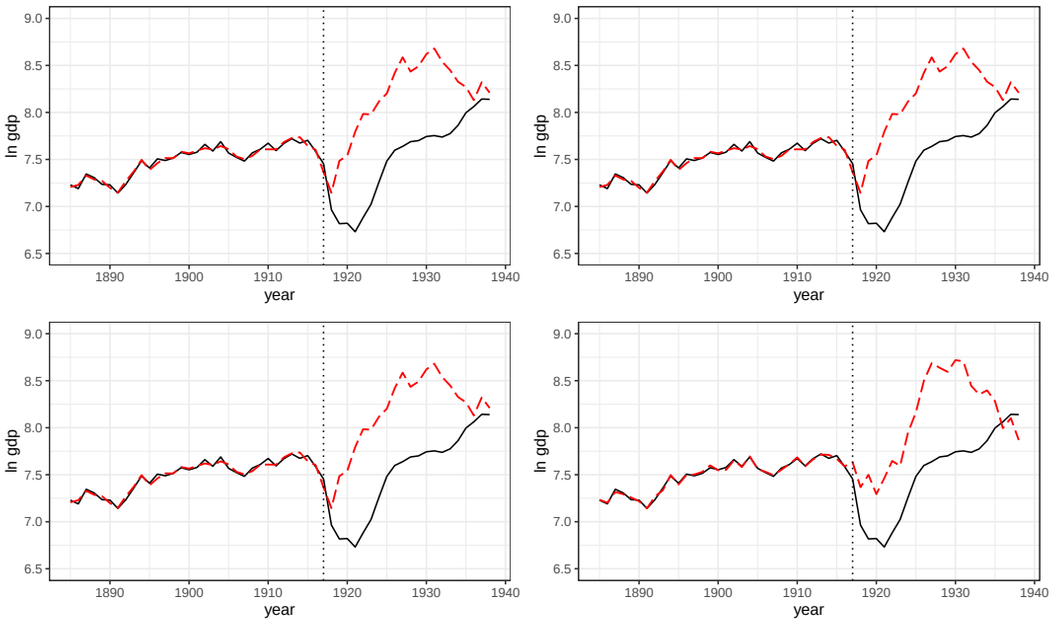
This section presents results from additional methods. First, Figure 10 presents the results from the HCW method. The upper panel allows for up to $J_{max} = 8$ control variables,

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the bottom panel allows for up to $J_{max} = 28$ regressors. The left panel uses the AICc to select the optimal model, the right panel uses the BIC. Notably, the HCW method fits Russian log GDP per capita very closely before the Revolution. This feature is not unique to Russia. Figure 11 presents the log GDP per capita gaps for Russia and 30 placebo units from the HCW method, as well as the log GDP gaps from the linear trend model. Compared to the gaps from the synthetic control method in Figure 3, the gaps from the HCW method before 1917 are much smaller, meaning that it fits the pre-1917 data better for most countries. However, while the post-1917 placebo gaps from the synthetic control method tend to diverge gradually, the placebo gaps from the HCW method involve large sudden increases and decreases right after 1917, and overall appear to be more dispersed.

It appears that the HCW method suffers from overfitting in this application. Because the pre-treatment period is short (32 observations) and the number of possible controls is large (30 countries), the HCW approach tends to overfit the data in the pre-treatment period, especially when up to 28 regressors can be included in the model. Restricting maximum the number of regressors to 8 alleviates this problem slightly, but does not solve it completely. As is well known in statistics, methods that overfit in-sample tend to have large variance and may produce poor out-of-sample predictions. Not surprisingly, according to the results of the placebo test, out-of-sample predictions from the HCW method appear to be unreliable.

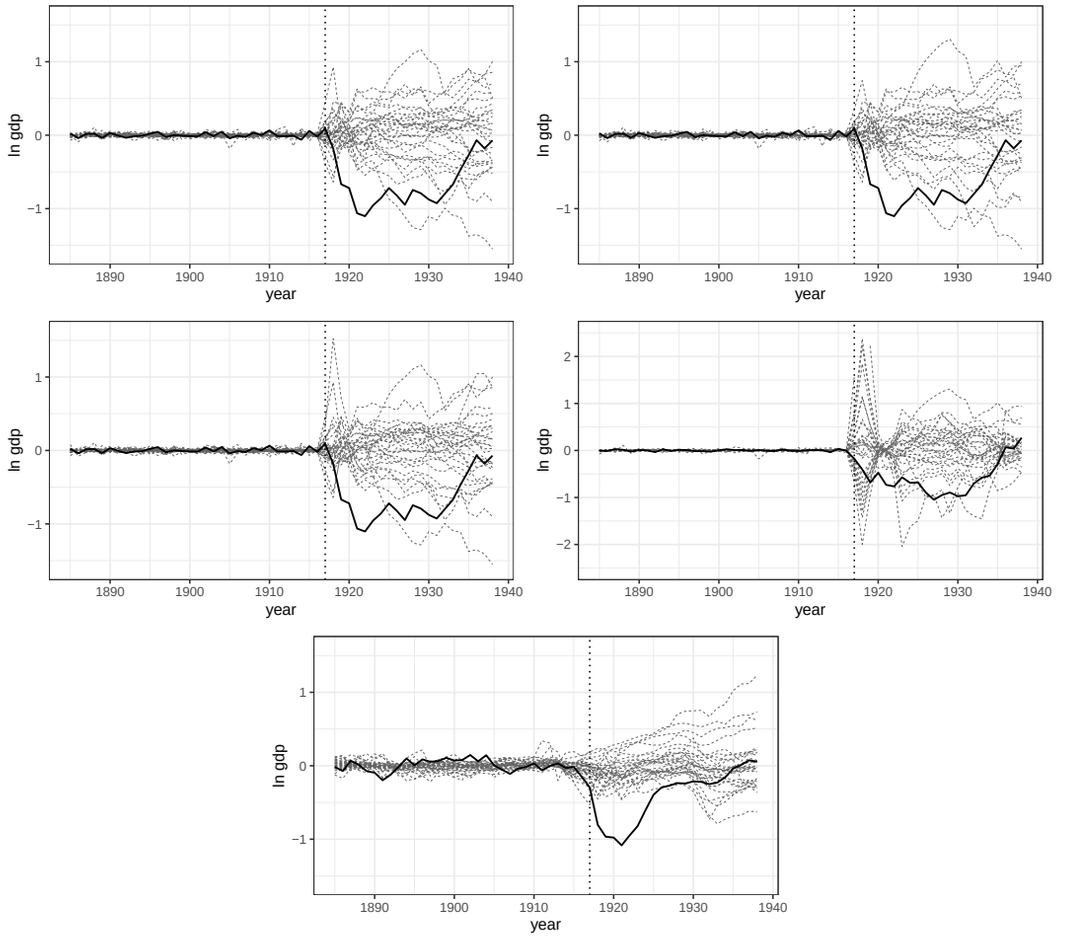
Figure 10: Actual and Counterfactual Log GDP per Capita, HCW Method



Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. Top left: the HCW method with up to 8 regressors and AICc. Top right: the HCW method with up to 8 regressors and BIC. Bottom left: the HCW method with up to 28 regressors and AICc. Bottom right: the HCW method with up to 28 regressors and BIC.

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Figure 11: Log GDP Gaps, Russia and Placebo Units, HCW Method



Notes: Black solid line – difference between the actual and counterfactual log GDP per capita for Russia. Gray dashed lines – differences between the actual and counterfactual log GDP per capita for 30 placebo units. Top left: the HCW method with up to 8 regressors and AICc. Top right: the HCW method with up to 8 regressors and BIC. Middle left: the HCW method with up to 28 regressors and AICc. Middle right: the HCW method with up to 28 regressors and BIC. Bottom: linear time trend (as baseline).

In particular, the counterfactuals for Russia based on the HCW method predict very rapid economic growth during the 1920s, but then a large decrease in GDP per capita during the 1930s. These counterfactuals look overly volatile and somewhat implausible. Table VI, which presents the estimated regression coefficients for the HCW method, may help understand where the problem comes from. The coefficients in three out of four specifications sum up to roughly to 2, and in the remaining specification, they sum up to almost 3. Thus, if all countries in the donor pool had grown at the same rate, the counterfactual unit would have grown 2–3 times faster.

However, when the donor pool is restricted only to the countries that suffered heavy World War I losses, the counterfactual, shown in Figure 12, becomes more believable. Because there are now fewer available controls, the HCW method no longer overfits before 1917. It predicts economic decline as a result of World War I, but then recovery and growth during the 1920s and most of the 1930s. As compared to the synthetic control method, these counterfactuals are more optimistic.

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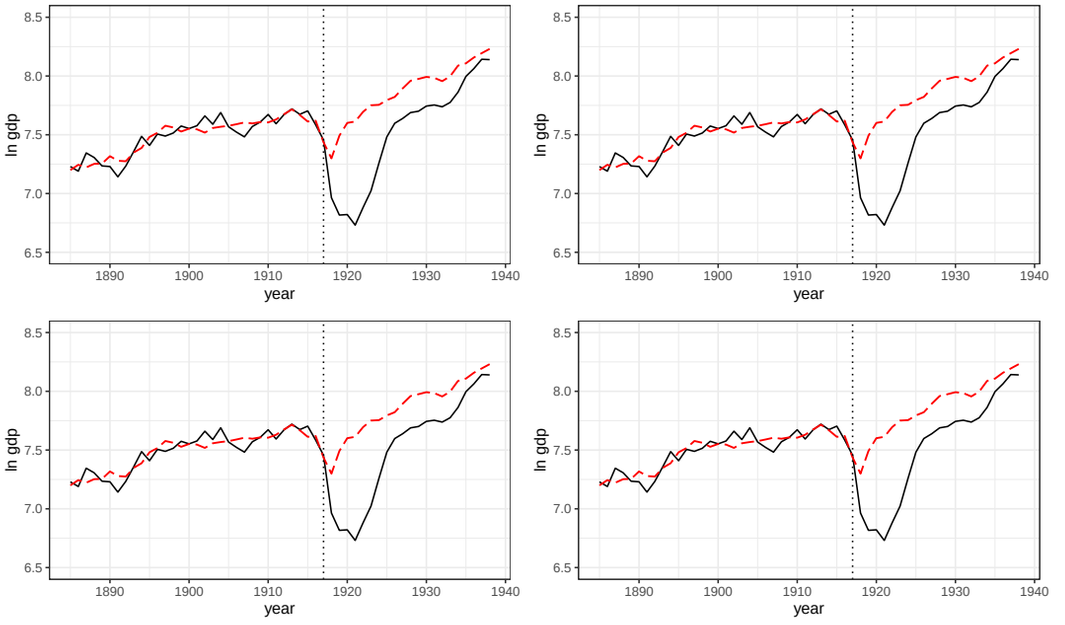
TABLE VI

HCW WEIGHTS

	AICc, $J_{max} = 8$	BIC, $J_{max} = 8$	AICc, $J_{max} = 28$	BIC, $J_{max} = 8$
Austria	-	-	-	-0.787
Brazil	-0.697	-0.697	-0.697	-0.622
Chile	-	-	-	0.348
Denmark	0.847	0.847	0.847	0.784
France	-	-	-	-0.566
Netherlands	2.379	2.379	2.379	3.343
New Zealand	-0.571	-0.571	-0.571	-0.729
Norway	-1.131	-1.131	-1.131	-3.063
Peru	-0.423	-0.423	-0.423	-0.450
Portugal	0.875	0.875	0.875	-
Spain	-	-	-	0.386
SriLanka	-	-	-	0.923
Sweden	0.820	0.820	0.820	1.645
Switzerland	-	-	-	0.762
UK	-	-	-	1.159
Uruguay	-	-	-	0.362
USA	-	-	-	-0.497
Total	2.099	2.099	2.099	2.998

Notes: The table presents the regression coefficients in various specifications of the HCW method.

Figure 12: Actual and Counterfactual Log GDP per Capita,, HCW Method, WWI



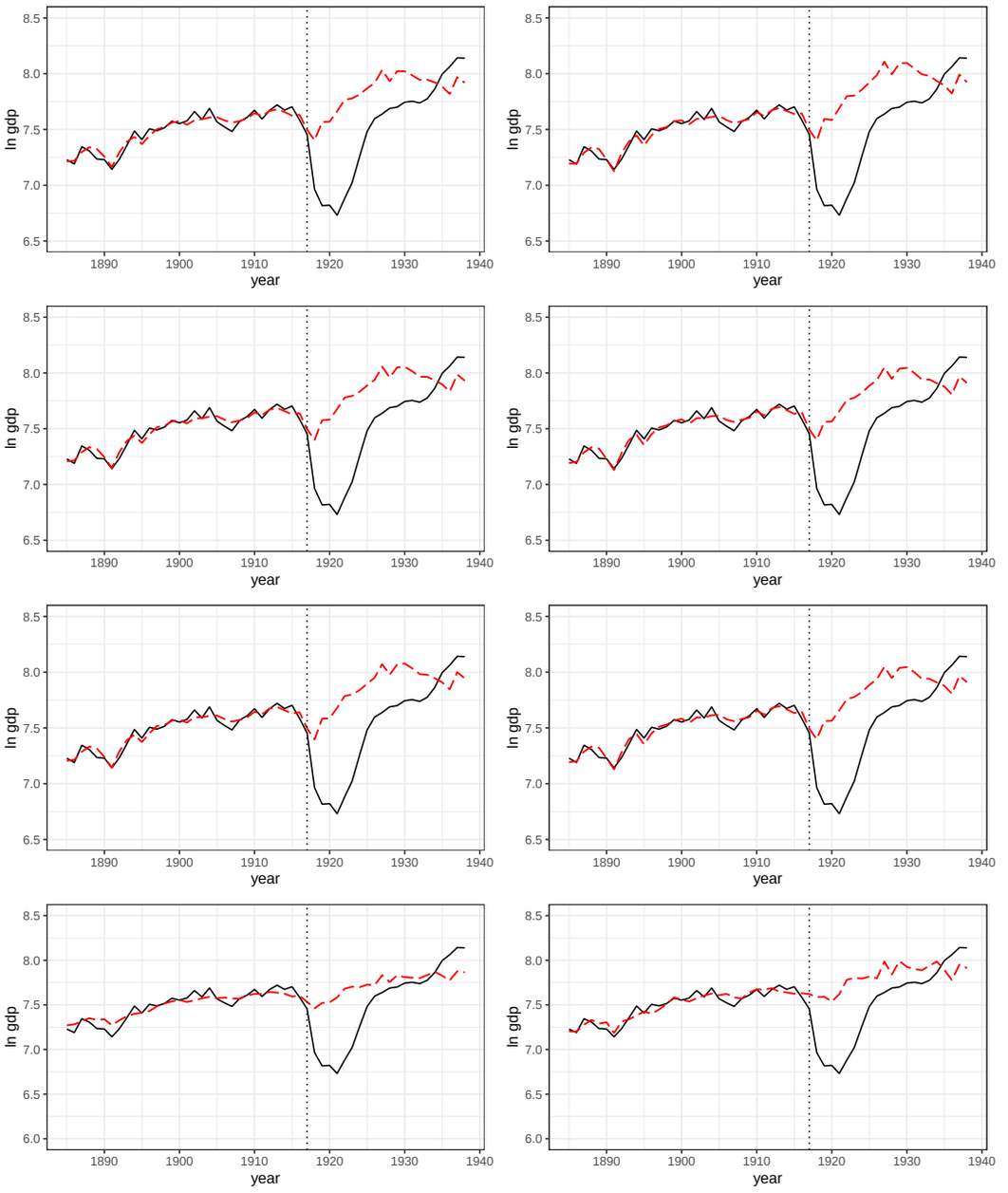
Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. Top left: the HCW method with up to 8 regressors and AICc. Top right: the HCW method with up to 8 regressors and BIC. Bottom left: the HCW method with up to 9 regressors and AICc. Bottom right: the HCW method with up to 9 regressors and BIC. Only the countries that lost at least 0.5% of their population in World War I in the donor pool in all specifications.

Figure 13 presents the results from the lasso and post-lasso OLS. The upper panel selects the tuning parameter using leave-one-out cross-validation (LOOCV), the second

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panel uses 4-fold cross-validation, the third panel uses 8-fold cross-validation, and the bottom panel uses placebo-based cross-validation. The left panel uses the lasso, the right panel uses the post-lasso OLS. In turn, Figure 14 presents the log GDP per capita gaps for Russia and 30 placebo units from the the lasso and post-lasso OLS.

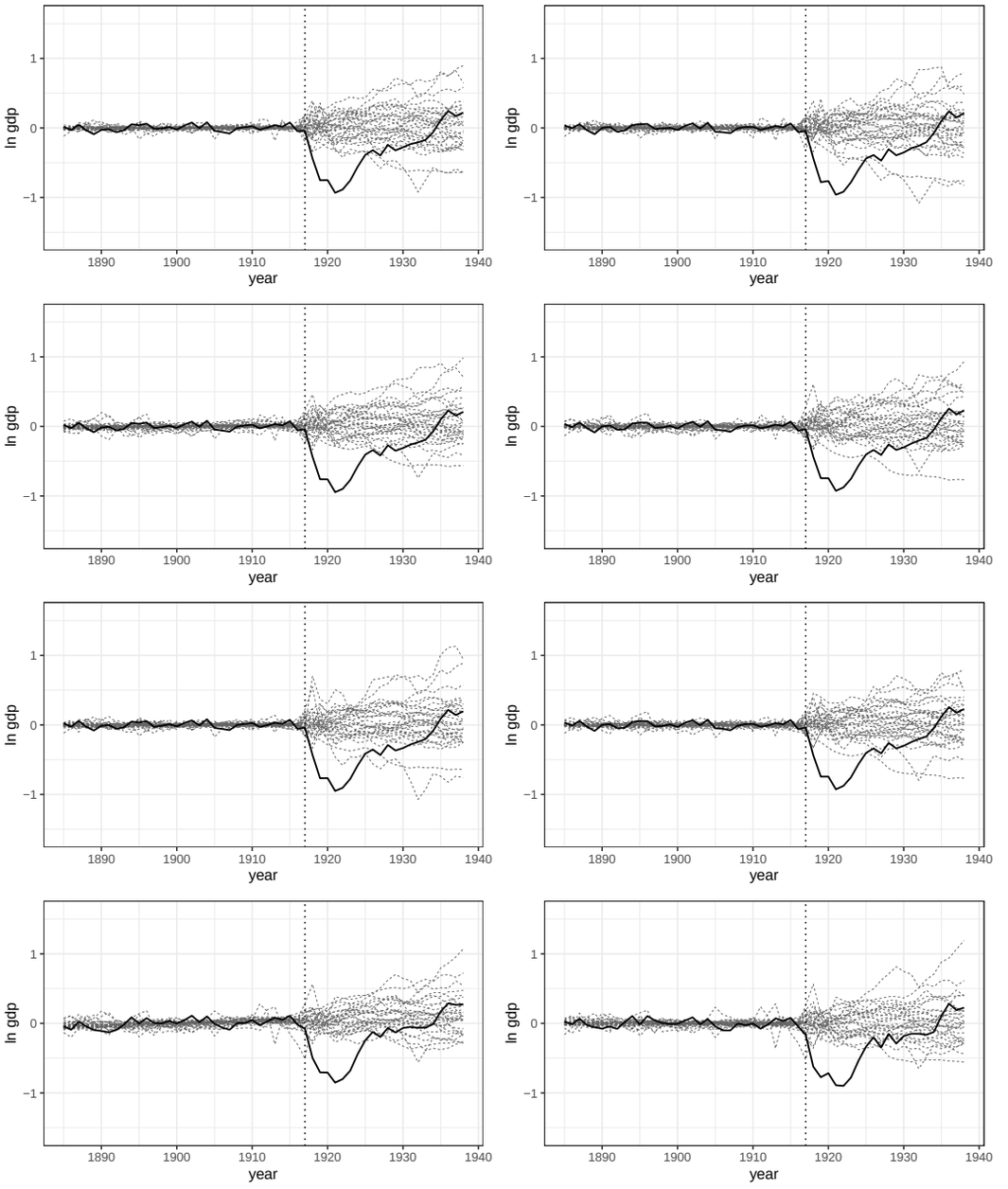
Figure 13: Actual and Counterfactual Log GDP per Capita, Lasso and Post-Lasso



Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. First row, left: the lasso with leave-one-out cross-validation (LOOCV). First row, right: post-lasso OLS with LOOCV. Second row, left: the lasso with 4-fold cross-validation. Second row, right: post-lasso OLS with 4-fold cross-validation. Third row, left: the lasso with 8-fold cross-validation. Third row, right: post-lasso OLS with 8-fold cross-validation. Last row, left: the lasso with placebo-based cross-validation. Last row, right: post-lasso OLS with placebo-based cross-validation.

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Figure 14: Log GDP Gaps, Russia and Placebo Units, Lasso and Post-Lasso



Notes: Black solid line – difference between the actual and counterfactual log GDP per capita for Russia. Gray dashed lines – differences between the actual and counterfactual log GDP per capita for 30 placebo units. First row, left: the lasso with leave-one-out cross-validation (LOOCV). First row, right: post-lasso OLS with LOOCV. Second row, left: the lasso with 4-fold cross-validation. Second row, right: post-lasso OLS with 4-fold cross-validation. Third row, left: the lasso with 8-fold cross-validation. Third row, right: post-lasso OLS with 8-fold cross-validation. Last row, left: the lasso with placebo-based cross-validation. Last row, right: post-lasso OLS with placebo-based cross-validation.

The counterfactual from the lasso-based methods is less extreme than from the HCW method, but qualitatively its behavior is similar. In particular, the lasso-based methods with LOOCV or k-fold cross-validation fits Russian log GDP per capita before the Revolution very closely, though less so than the HCW method. Likewise, as we can see from Figure 14, the lasso-based methods generally result is small GDP per capita gaps before

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1917 for most countries. It appears that the lasso-based methods with LOOCV and k-fold cross-validation also suffer from overfitting, because the chosen values of the tuning parameter λ are too low and fail to prevent overfitting.

In general, leave-one-out cross-validation tends to work well when the observations are independent, but it may fail when there is dependence between observations. Since the left-out observation is highly correlated with the preceding and subsequent observations, fitting those observations well will also result in a good fit of the left-out observation. Because of this, leave-one-out cross-validation may underestimate the true out-of-sample MSE and select the tuning parameter that overfits the data. In principle, k-fold cross-validation based on the blocks of data could alleviate this problem, but in this case it does not. In contrast, the lasso and post-lasso OLS with placebo-based cross-validation do not seem to suffer from overfitting, because the selected values of the tuning parameter λ are higher. As a result, the counterfactuals based on these methods are more moderate and less volatile.

Figure 15 presents the results from the lasso and post-lasso OLS when the donor pool is restricted to World War I. Similarly to the HCW method above, restricting the donor pool helps avoid overfitting and results in more plausible counterfactuals that are largely in line with the counterfactuals based on the synthetic control method.

Table VII displays the estimated coefficients ("weights") for the lasso and post-lasso OLS. I do not report the intercept term as it does not have a substantive meaning. This table helps understand the source of overfitting and unrealistic out-of-sample forecasts. We can see that the lasso and post-lasso select a large number of controls, and the weights in most specifications sum up roughly to 1.5. Thus, if all countries in the donor pool had grown at the same rate, the counterfactual unit would have grown 1.5 times faster. This might explain why the counterfactuals based on the lasso and post-lasso exhibit a very fast economic growth during the 1920s.

Table VIII presents the lasso and post-lasso OLS weights when the donor pool is restricted to World War I. Unlike in Table VII, the sum of these weights happens to be close to 1. This may explain why the counterfactuals based on the restricted donor pool are less volatile and are more similar to the counterfactuals based on the synthetic control method.

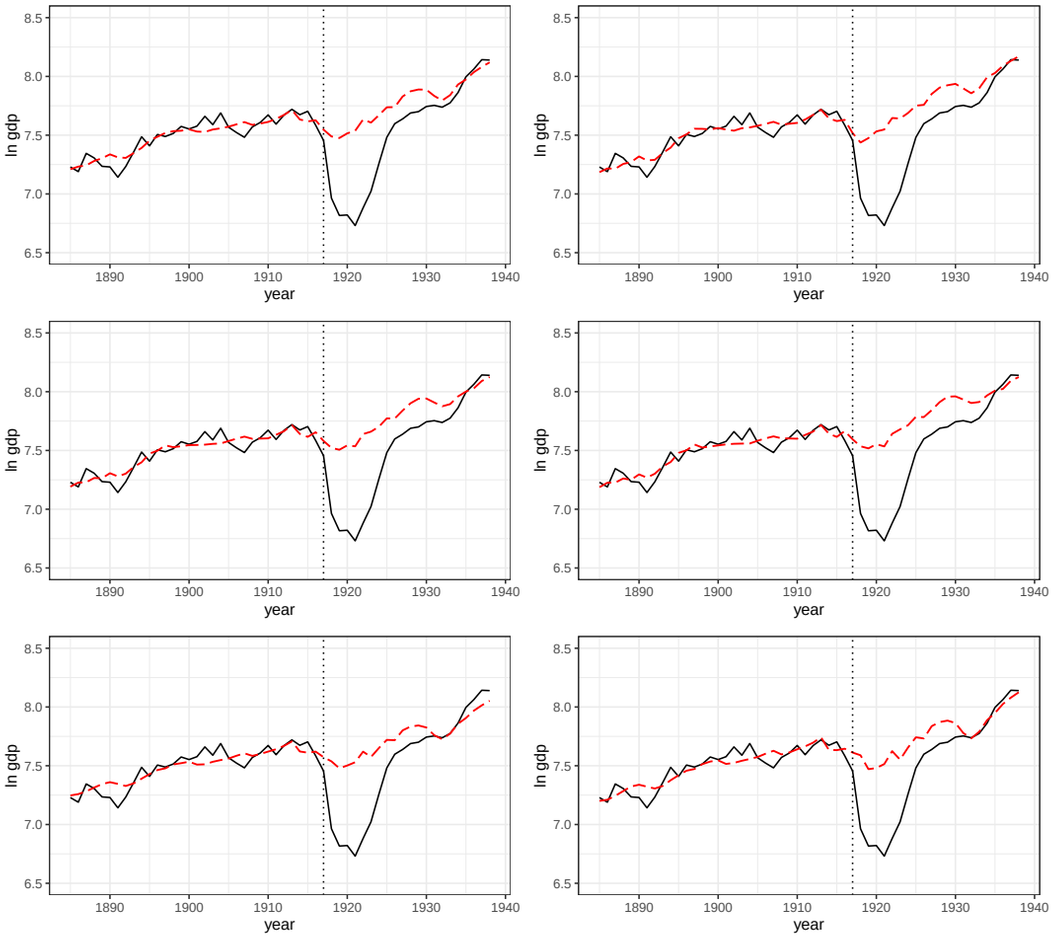
4.4. *Discussion*

Several remarks are in order to better put my results into context. First, as discussed by Markevich and Harrison (2011, p. 691), it is probably impossible to explain why the Revolution took place in Russia and not elsewhere "without reference to historical factors that were unique to that country and period." Because of this, my estimates should be viewed as predictive rather than causal. The Revolution and the Civil War might have happened in Russia because of certain country-specific factors, and these factors might have affected Russian economic growth even if the Revolution had not happened. The methods I use may fail to account for this. Thus, I cannot determine whether the difference between the actual and counterfactual unit should be attributed to the Revolution itself, the Civil War, consequences of World War I, or some preexisting factors specific to Russia.

Second, the relationship that I estimate can be viewed as a "reduced form" relationship. This distinguishes my approach from the previous simulation studies, such as Hunter and Szyrmer (2014), Allen (2003), and Cheremukhin, Golosov, Guriev, and Tsyvinski (2017). For instance, Hunter and Szyrmer (2014) use the KAPROST model that centers on the process of enlarging the capital stock. They consider both a disaggregated model

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Figure 15: Actual and Counterfactual Log GDP per Capita, Lasso and Post-Lasso, WWI



Notes: Black solid line – actual log GDP per capita. Red dashed line – counterfactual log GDP per capita. First row, left: the lasso with leave-one-out cross-validation (LOOCV). First row, right: post-lasso OLS with LOOCV. Second row, left: the lasso with 4-fold cross-validation. Second row, right: post-lasso OLS with 4-fold cross-validation. Third row, left: the lasso with 8-fold cross-validation. Third row, right: post-lasso OLS with 8-fold cross-validation. Only the countries that lost at least 0.5% of their population in World War I in the donor pool in all specifications.

with 12 different producing sectors and 7 final demand categories and an aggregated one-sector model. Based on their model, they construct various counterfactuals for 1928–1940 and conclude that Russia, or the Soviet Union, could have grown faster under alternative policies. More specifically, they argue that the policies implemented by the Soviet government in the 1930s went too far and were too demanding. For instance, the Soviet economy might have performed better without collectivization.

Allen (2003) also uses a model that focuses on capital stock accumulation to construct a counterfactual for 1928–1940. However, his model has three sectors: agriculture, consumer goods, and producer goods. He assumes that there is "excess population" in agriculture, so labor supply does not constrain the output. In contrast to the results of Hunter and Szyrmer (2014), Allen (2003) finds that the counterfactual capitalist economy would have performed worse than the actual Soviet economy. Moreover, if the government had decided to stick to the New Economic Policy, the output would have been lower. Thus, his results indicate that the Soviet economy did fairly well in the 1930s relative to various

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TABLE VII

LASSO AND POST-LASSO WEIGHTS

	LOOCV		4-fold CV		8-fold CV		Placebo-Based CV	
	Lasso	Post-L	Lasso	Post-L	Lasso	Post-L	Lasso	Post-L
Argentina	0.164	0.186	0.157	0.138	0.148	0.138	0.165	0.482
Australia	-	-	-0.053	-	-0.076	-	-	-
Brazil	-0.571	-0.699	-0.589	-0.750	-0.590	-0.750	-0.129	-
Chile	0.009	-	0.025	0.063	0.023	0.063	-	-
Finland	-	-0.118	-	-	-	-	0.135	-0.414
India	0.229	0.214	0.226	0.189	0.224	0.189	0.326	0.684
Indonesia	-	-	-	-	0.006	-	-	-
Netherlands	0.848	1.158	0.942	1.003	0.984	1.003	0.022	-
Portugal	0.795	0.857	0.798	0.817	0.796	0.817	0.649	1.210
Sri Lanka	0.038	0.077	0.011	-0.013	0.018	-0.013	-	0.079
UK	-	-	-	-	-	-	0.127	0.332
Uruguay	-	-	-	-	0.014	-	-	-
Venezuela	-0.001	0.011	-0.005	0.010	-0.004	0.010	-0.047	-0.016
Total	1.510	1.687	1.512	1.457	1.543	1.457	1.247	2.357

Notes: The table presents the lasso coefficients ("weights") in various specifications. LOOCV refers to the use of leave-one-out cross-validation for the tuning parameter choice. 4-fold CV refers to the use of 4-fold cross-validation based on 4 blocks of data for the tuning parameter choice. 8-fold CV refers to the use of 8-fold cross-validation based on 8 blocks of data for the tuning parameter choice. Placebo-Based CV refers to the use of placebo-based cross-validation, described in the text.

possible counterfactual scenarios.

Cheremukhin, Golosov, Guriev, and Tsyvinski (2017) study the performance of the Tsarist Russian economy as well as the Soviet economy in 1928–1940. They consider a neoclassical two-sector growth model and focus on frictions, or wedges, that prevent the economy from functioning efficiently. They argue that entry barriers and monopoly power were the primary causes for Russia's failure to industrialize, while the Soviet economic growth of the 1930s was achieved primarily through reducing these frictions despite lower productivity in both sectors. They also construct two counterfactuals. The first one continues the path of the Tsarist economy up to 1940 assuming that the sectoral TFPs would follow their pre-1913 log-linear trends and all distortions would remain the same. The Soviet GDP per capita in 1939 is higher than in this counterfactual by around 6%. On the other hand, the counterfactual in which the labor and capital wedges are set to zero, while all other components are kept at their 1913 levels, significantly outperforms the actual Soviet economy.

Unlike these three studies, I do not attempt to develop a "structural model" of the Russian or Soviet economy. As a result, I cannot study the effect of individual policies on Russian economic growth; I can only obtain a single counterfactual that combines all policies together. On the other hand, I am able to obtain the counterfactual not only for 1928–1938, as in all these studies, but also for 1917–1927. As far as the conclusions about the performance of the Soviet economy go, my results are somewhere in between Allen (2003), who views the Soviet economy of the 1930s favorably, and Hunter and Szyrmer (2014), who argue that it could have done better.

More specifically, most of my preferred counterfactuals agree with Allen (2003), and with the Tsarist counterfactual with fixed wedges in Cheremukhin, Golosov, Guriev, and

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TABLE VIII

LASSO AND POST-LASSO WEIGHTS, WWI

	LOOCV		4-fold CV		8-fold CV	
	Lasso	Post-L	Lasso	Post-L	Lasso	Post-L
Australia	-0.159	-0.293	-0.353	-0.417	-	-
Finland	0.462	0.653	0.450	0.439	0.265	0.183
Germany	0.319	0.242	0.198	0.153	0.382	0.522
Italy	-	-	0.381	0.524	-	-
UK	0.411	0.358	0.391	0.382	0.434	0.624
Total	1.033	0.959	1.067	1.080	1.080	1.329

Notes: The table presents the lasso coefficients ("weights") in various specifications when only countries that lost at least 0.5% of their population in World War I (Austria, Finland, France, Germany, Italy, the United Kingdom, Australia, New Zealand, and Canada) are included in the donor pool. LOOCV refers to the use of leave-one-out cross-validation for the tuning parameter choice. 4-fold CV refers to the use of 4-fold cross-validation based on 4 blocks of data for the tuning parameter choice. 8-fold CV refers to the use of 8-fold cross-validation based on 8 blocks of data for the tuning parameter choice. Placebo-Based CV refers to the use of placebo-based cross-validation, described in the text.

Tsyvinski (2017), in that it predicts that Russia would have been less developed without the Revolution by the end of the 1930s than the Soviet Union actually was. At the same time, I should emphasize that the uncertainty about the counterfactuals for the end of the 1930s is inherently very high, and there are alternative specifications where the counterfactual does as well by the end of the 1930s as the actual Soviet Union. Moreover, while the counterfactual is typically more affected by the Great Depression in the 1930s, it does not experience the dramatic economic downturn associated with the Civil War around 1920. I should also note the Soviet policies of the 1930s were associated with the disastrous famine of 1932–1933 and the repressions of the late 1930s, during which several million people died. For example, Naumenko (2021) shows that in Ukraine, where 2.6 million people died from the famine, over half of excess deaths can be attributed to government policies, such as collectivization. I abstract away from any welfare calculations in this paper, but any such calculations would need to take a stand on the implications of various policies in terms of lives lost and their respective economic costs.

Finally, in order for my counterfactual to be credible, I need to assume that the predictive relationship that I estimate would have remained unchanged without the Revolution. This assumption may be problematic. For instance, as noted by Davies (1994), the capital goods industries in Tsarist Russia were largely foreign-owned. Moreover, Russia was one of world leaders in grain production and export (see, e.g. Falkus (1966); Gregory (2014)). As such, Russia might have been especially reliant on the world economy and international trade in certain ways. Given that World War I resulted in the disintegration of the world economy and a major reduction in international trade (Glick and Taylor (2010); Federico and Tena-Junguito (2019)), these factors might have affected Russia more negatively than most countries in the donor pool. I could try to account for them by including countries that were similar to Russia in the donor pool, but, as discussed above, the data for other Eastern European countries for 1885–1938 is not available.

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5. CONCLUSION

In this paper I use different econometric methods from the recent literature on program evaluation, such as the HCW method, the lasso, and the synthetic control method, to predict how Russia could have developed without the Revolution of 1917. I run a number of placebo tests to compare different methods and determine that the synthetic control method, and especially its demeaned version, demonstrate the best performance. As the number of potential controls is almost as large as the number of pre-1917 observations, other approaches seem to suffer from overfitting. In contrast, the synthetic control method appears to impose enough restrictions to avoid overfitting.

I perform a number of robustness checks and find that the counterfactual Russian growth path for the 1920s is fairly robust to the modeling choices that I make. While I abstract away from sampling uncertainty and inference in this paper, it appears that without the Revolution, Russia might have experienced a relatively short downturn as a result of World War I but would have recovered fairly quickly and would have grown steadily throughout the 1920s.

The counterfactuals for the 1930s exhibit more variation. Some of the counterfactuals predict that without the Revolution, Russia might have reached similar levels of economic development as the Soviet Union. Others suggest that Russia would have experienced a prolonged economic downturn during the 1930s and that the Soviet Union would have overtaken the counterfactual by the mid-1930s.

As my placebo tests show, the uncertainty about the predictions from the synthetic control method increases as the forecasting horizon becomes longer. Thus, it is difficult to conclusively answer the question of whether the Soviet Union did better or worse in the 1930s than Russia would have done without the Revolution. However, it does appear that the Revolution and the Civil War were associated with large economic losses during the 1920s, as compared to the counterfactual.

APPENDIX A: ADDITIONAL RESULTS

A.1. *Modeling in Levels, Logs, or Growth Rates*

In this section I discuss whether I should model the outcome variable in levels, logarithms, or in growth rates (or, equivalently, first differences in logarithms). There is no agreement in the literature on the choice of the outcome variable. For instance Hsiao, Ching, and Wan (2012) and Gardeazabal and Vega-Bayo (2017) focus on modeling growth rates. In fact, Hsiao, Ching, and Wan (2012) assume stationarity in their analysis, which clearly rules out working with levels (or logs), because GDP per capita increases over time. However, Bai, Li, and Ouyang (2014) show that the HCW approach can be used even when the data are non-stationary.

In contrast, many papers in the synthetic control literature, such as Abadie and Gardeazabal (2003) and Abadie, Diamond, and Hainmueller (2015), model GDP per capita in levels instead of growth rates. Finally, there are also papers, e.g. Gardeazabal and Vega-Bayo (2016), that apply the synthetic control method to the logarithm of GDP per capita. Time series econometricians typically prefer to use the logarithm of GDP per capita as a dependent variable, because economists often believe that growth rates are stationary, and hence the first differences of the logarithm of GDP per capita are stationary as well. I follow the tradition in the time series literature and use the logarithm of GDP per capita, rather than its level. However, I still need to decide whether to use the logarithm of GDP as an outcome variable directly, or to use its first difference (i.e. the GDP per capita growth rate) as an outcome variable and then reconstruct the logarithm series from first differences.

While I am primarily interested in GDP per capita (or its logarithm) rather than in its growth rate, I turn to statistics to choose the appropriate outcome variable. More specifically, I evaluate which choice allows me to obtain a good fit of both the logarithm series and GDP per capita growth rates in the pre-treatment period.

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I consider six methods in total: two versions of the HCW approach (best subset selection with the BIC and best subset selection with the AICc, with $J_{max} = 28$ in both cases), the lasso and post-lasso OLS, and two versions of the synthetic control method (the standard and demeaned one). For every method I use, I first estimate the model in logs and directly obtain the fitted values $\hat{y}_{1,t}^{logs}$. Then I construct the first differences from these fitted values: $\Delta\hat{y}_{1,t}^{logs} = \hat{y}_{1,t}^{logs} - \hat{y}_{1,t-1}^{logs}$. Next, I go the other way around: I estimate the model in the first differences (in logs), directly obtain the fitted values $\Delta\hat{y}_{1,t}^{diffs}$, and then reconstruct the series in logs as $\hat{y}_{1,t}^{diffs} = \hat{y}_{1,t-1}^{diffs} + \Delta\hat{y}_{1,t}^{diffs}$, where I let $\hat{y}_{1,1}^{diffs} = y_{1,1}$ (i.e. I use the actual value of the logarithm of GDP per capita in 1885 as the starting value).

Then I measure the fit of each counterfactual series by computing the mean squared error over the period between 1885 (or 1886 for first differences) and 1916: $MSE^{*} = \frac{1}{T_0}\sum_{t=1}^{32}(y_{1,t} - \hat{y}_{1,t}^{*})^2$ and $MSE_{\Delta}^{*} = \frac{1}{T_0-1}\sum_{t=2}^{32}(\Delta y_{1,t} - \Delta\hat{y}_{1,t}^{*})^2$, where $* = logs$ or $* = diffs$. MSE^{logs} and MSE_{Δ}^{diffs} measure the quality of "direct" fit when modeling is done in logs and first differences correspondingly. In turn, MSE_{Δ}^{logs} and MSE^{diffs} measure the quality of "indirect" fit, when modeling is done for one choice of the outcome variable but the MSE is computed for another.

Table A2 shows the resulting MSE, multiplied by 1,000, for the six methods I consider: the synthetic control method and its demeaned version, the HCW approach using the BIC and AICc to select the optimal number of regressors, the lasso, and post-lasso OLS. The first column corresponds to modeling in logs, the second column corresponds to modeling in first differences, and the last column shows the ratio of the first and second columns. Table A3 repeats the analysis for MSE_{Δ}.

We can make several observations based on these tables. First, estimating the model in differences using the HCW approach coupled with the BIC results in severe overfitting. The resulting model fits both first differences and logs series almost perfectly, leading to the very low values of the MSE for both outcome variables. At the same time, as we can see from Figures A2 and A3, the corresponding out-of-sample predictions are unrealistically volatile.

Next, while it is natural to expect that "direct" modeling should produce a better fit for a given outcome variable than "indirect" modeling, both lasso and post-lasso fit *both* outcome variables better when modeling is done in logs. The remaining three methods, the HCW approach with the AICc and both versions of the synthetic control method, yield expected results, where "direct" modeling is preferred to "indirect" one. However, the gains from using "direct" methods are asymmetric. Modeling in logs results in a significantly better fit of the logs outcome than modeling in first differences, reducing the MSE for logs by a factor of 4 to 11. In contrast, modeling in differences instead of logs for the differences outcome only improves the MSE for first differences by about 25 to 35% when the synthetic control method is used. For the lasso and post-lasso OLS, there is no substantial improvement at all.

Next, Figure A3 shows that while most of the estimates are "oversmoothed," i.e. less volatile than the actual series, none of the methods seems to exhibit any systematic bias for the differences outcome. In contrast, Figure A2 indicates that the lasso and both versions of the synthetic control method are severely biased downwards relative to the actual logs series when modeling is done in first differences. Intuitively, when modeling is done in differences and then the logs series is reconstructed, it behaves as a random walk: all past errors enter the current fitted value, and past errors never disappear. As a result, the errors may add up over time, leading to poor fit and bias.

Overall, it appears that modeling in logs is preferred to modeling in first differences: it substantially improves the fit of the logs series and only slightly reduces the quality of fit of the differences series. Thus, in the main text, all modeling is done in logarithms.

A.2. *Modeling in Levels instead of Logs*

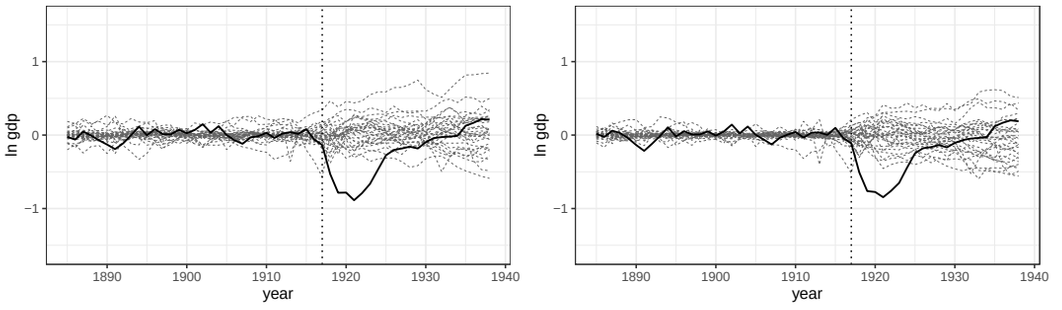
Finally, I study the sensitivity of my conclusions to the fact that I conduct my analysis in logs instead of levels. I repeat the analysis with the synthetic control method using the level of GDP per capita, instead of its natural logarithm, as the outcome variable. Figure A4 presents the results. The results remain qualitatively similar to those presented in the main text, though the counterfactual from the demeaned version of the synthetic control method, especially when the donor pool is restricted to the countries with heavy World War I losses, may seem overly optimistic.

Figure A5 plots the GDP per capita gaps for Russia and 30 placebo units. It seems that the synthetic control method does not do as well when modeling is done in levels as it does when modeling is done in logs.

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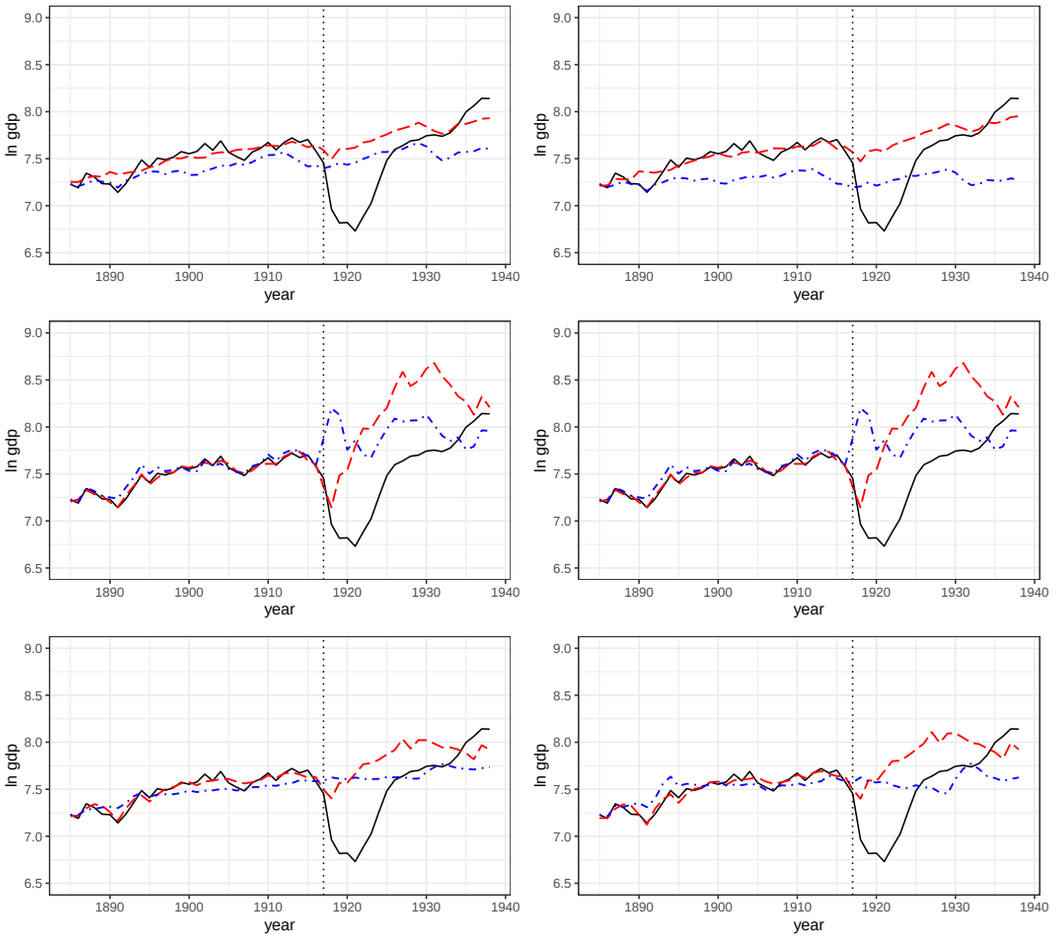
Table A4 presents the synthetic control weights when modeling is done in levels. Table A5 shows the balancing between the actual and the synthetic control units.

Figure A1: Log GDP Gaps, Russia and Placebo Units, Synthetic Control



Notes: Black solid line – difference between the actual and counterfactual log GDP per capita for Russia. Gray dashed lines – differences between the actual and counterfactual log GDP per capita for 30 placebo units. Left: synthetic control method. Right: synthetic control method, demeaned.

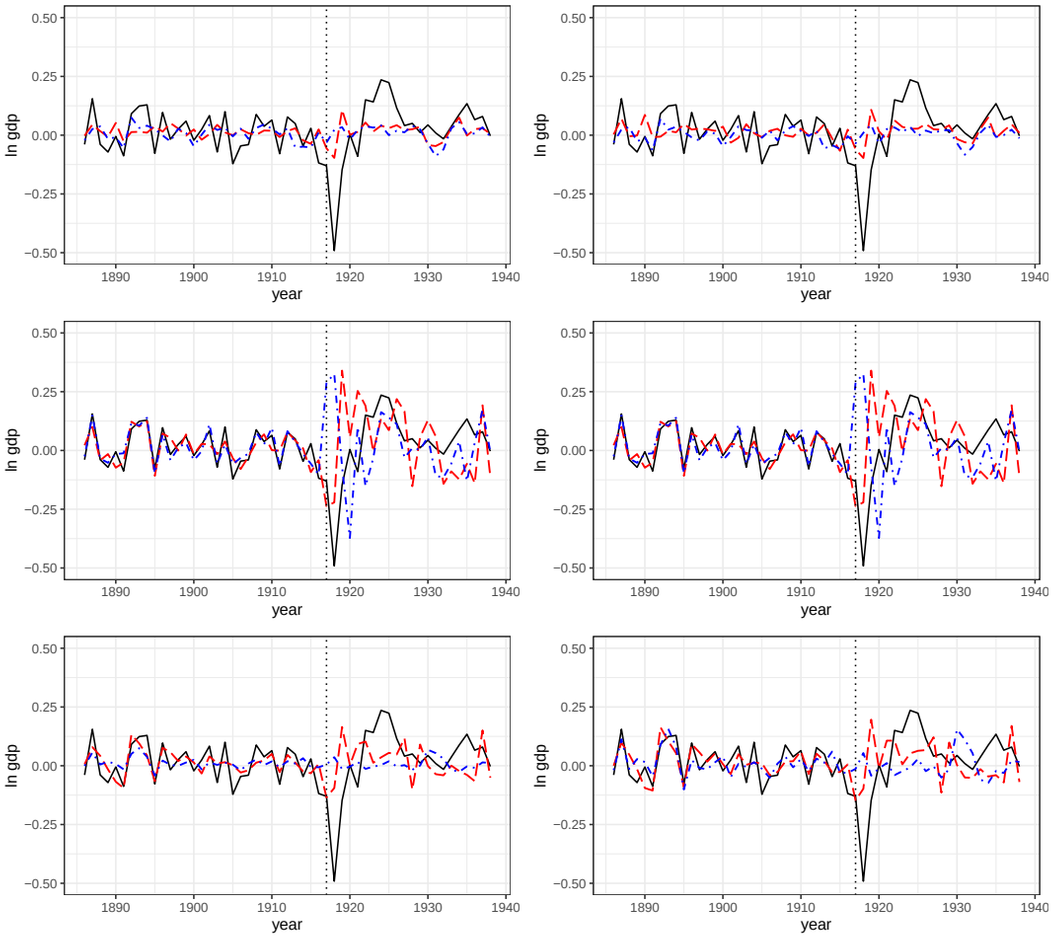
Figure A2: Actual and Counterfactual Log GDP per Capita, Modeling in Logs vs Growth Rates



Notes: The figure plots the actual log GDP per capita series (black solid line), the counterfactual log GDP per capita based on modeling in logs (red dashed line), and the counterfactual log GDP per capita based on modeling in first differences (blue dash-dotted line). Top left: synthetic control method. Top right: synthetic control method, demeaned. Middle left: HCW approach with BIC. Middle right: HCW approach with AICc. Bottom left: lasso. Bottom right: post-lasso.

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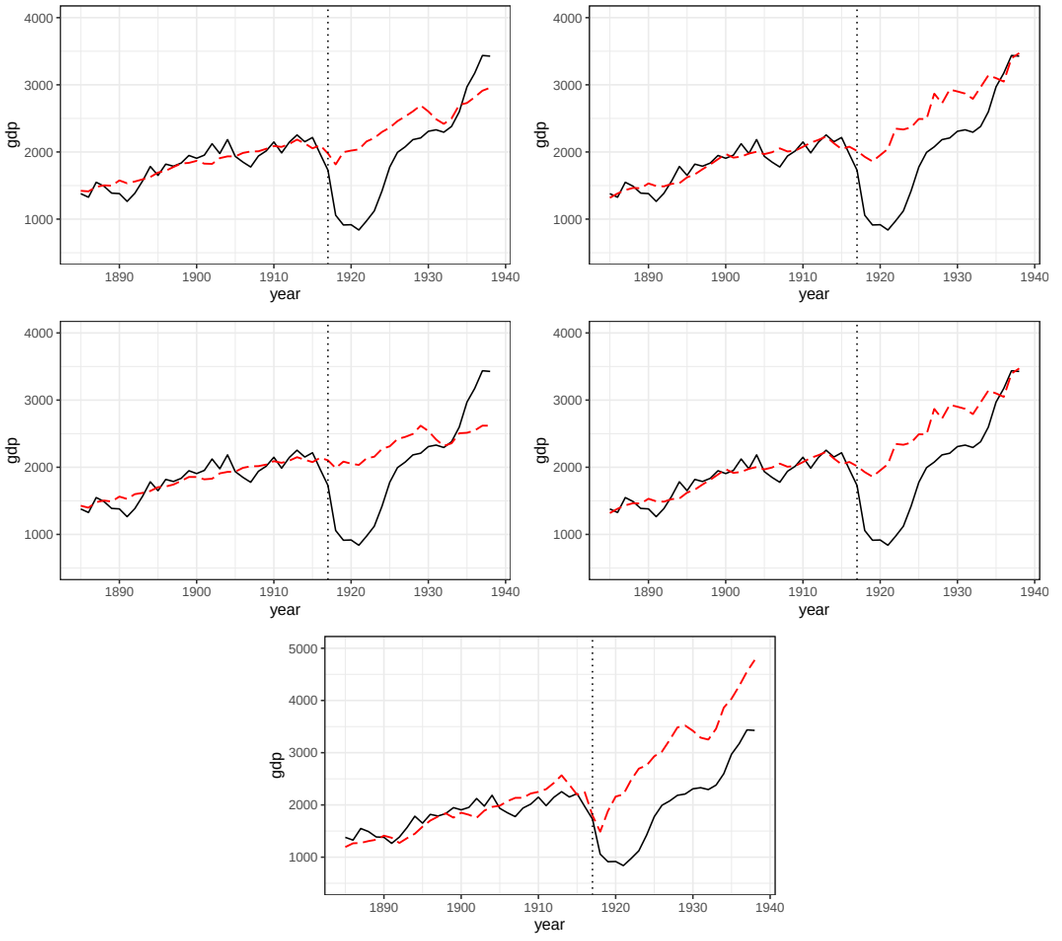
Figure A3: Actual and Counterfactual GDP per Capita Growth Rate, Modeling in Logs vs Growth Rates



Notes: The figure plots the actual log GDP per capita series (black solid line), the counterfactual log GDP per capita based on modeling in logs (red dashed line), and the counterfactual log GDP per capita based on modeling in first differences (blue dash-dotted line). Top left: synthetic control method. Top right: synthetic control method, demeaned. Middle left: HCW approach with BIC. Middle right: HCW approach with AICc. Bottom left: lasso. Bottom right: post-lasso.

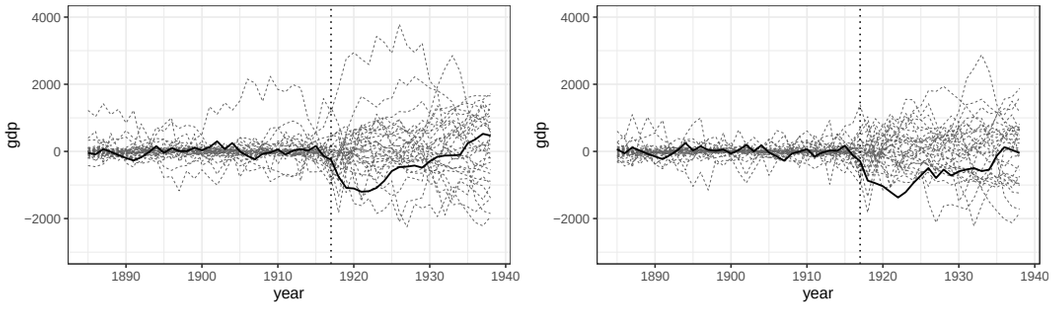
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Figure A4: Actual and Counterfactual GDP per Capita, Synthetic Control, Levels



Notes: Black solid line – actual GDP per capita. Red dashed line – counterfactual GDP per capita. Top left: synthetic control method. Top right: Top left: synthetic control method, demeaned. Middle left: synthetic control method, Finland is dropped from the donor pool. Middle right: synthetic control method, demeaned, Finland is dropped from the donor pool. Bottom: synthetic control method, demeaned, only World War I participants in the donor pool.

Figure A5: GDP Gaps, Russia and Placebo Units, Synthetic Control



Notes: Black solid line – difference between the actual and counterfactual GDP per capita for Russia. Gray dashed lines – differences between the actual and counterfactual GDP per capita for 30 placebo units. Left: synthetic control method. Right: synthetic control method, demeaned.

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TABLE A1

COUNTRIES

Country	Population, 1913 (thousands)	WWI Losses (thousands)	WWI Losses (% of population)
Russia	156,192	1,700.0	1.088
Austria	6,767	120.0#	1.773
Belgium	7,666	13.7	0.179
Denmark	2,983	0.0	0
Finland	3,027	26.5#	0.876
France	41,463	1,357.8	3.275
Germany	65,058	1,773.7	2.726
Italy	37,248	650.0	1.745
Netherlands	6,164	0.0	0
Norway	2,447	0.0	0
Sweden	5,621	0.0	0
Switzerland	3,864	0.0	0
UK	45,649	702.4*	1.539
Greece	5,425	5.0	0.092
Portugal	5,972	7.2	0.121
Spain	20,263	0.0	0
Australia	4,821	59.9*	1.242
New Zealand	1,122	16.7*	1.489
Canada	7,852	62.8*	0.800
USA	97,606	116.5	0.119
Argentina	7,653	0.0	0
Brazil	23,660	0.0	0
Chile	3,431	0.0	0
Colombia	5,195	0.0	0
Peru	4,295	0.0	0
Uruguay	1,177	0.0	0
Venezuela	2,874	0.0	0
India	303,700	54*	0.017
Indonesia	51,637	0.0	0
Japan	51,672	0.3	0.001
Sri Lanka	4,811	0.0	0

Notes: The data on the 1913 population is from the Maddison Project Database 2020 (`https://www.rug.nl/ggdc/historicaldevelopment/maddison/releases/maddison-project-database-2020`). "WWI Losses" takes into account military losses only and comes from Encyclopedia Britannica (`http://www.britannica.com/EBchecked/topic/648646/World-War-I/53172/Killed-wounded-and-missing`). For the countries marked by "*" (that were parts of the British Empire), the data is from Ellis and Cox (2001), except India, for which the data is from Urlanis (1971). Austria and Finland, marked by "#", did not exist in modern borders in 1914. For Austria, I use data from Erlikman (2004); for Finland, from International Labor Office (1923–1925).

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TABLE A2

$1,000 \times MSE$, 1885–1916

Outcome: Logs	Modeling: Logs	Modeling: Differences	Ratio
Synthetic	5.704	22.730	0.251
Synthetic, demeaned	5.570	63.810	0.087
HCW, AICc	0.937	2.775	0.338
HCW, BIC	0.937	2.775	0.338
Lasso	2.041	8.403	0.243
Post-Lasso	1.886	8.123	0.232

Notes: The table presents $MSE^* = \frac{1}{T_0}\sum_{t=1}^{32}(y_{1,t} - \hat{y}^*_{1,t})^2$, where $* = logs$ or $* = diffs$, multiplied by 1,000. See Section A.1 for details. The Ratio column is equal to the ratio of the first and second columns.

TABLE A3

$1,000 \times MSE_\Delta$, 1885–1916

Outcome: Differences	Modeling: Logs	Modeling: Differences	Ratio
Synthetic	6.015	4.841	1.242
Synthetic, demeaned	6.468	4.857	1.332
HCW, AICc	2.641	1.069	2.470
HCW, BIC	2.641	1.069	2.470
Lasso	3.403	4.103	0.829
Post-Lasso	3.394	3.350	1.013

Notes: The table presents $MSE^*_\Delta = \frac{1}{T_0-1}\sum_{t=2}^{32}(\Delta y_{1,t} - \Delta\hat{y}^*_{1,t})^2$, where $* = logs$ or $* = diffs$, multiplied by 1,000. See Section A.1 for details. The Ratio column is equal to the ratio of the first and second columns.

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TABLE A4

SYNTHETIC CONTROL WEIGHTS, LEVELS

Country	Baseline		No Finland		WWI	
	Synthetic	Synthetic (demeaned)	Synthetic	Synthetic (demeaned)	Synthetic	Synthetic (demeaned)
Argentina	0.034	-	0.058	-	-	-
Canada	-	-	0.005	-	-	-
Chile	-	-	0.001	-	-	-
Denmark	-	0.092	0.005	0.092	-	-
Finland	0.225	-	-	-	1	0.896
Germany	-	0.141	0.002	0.141	-	-
India	0.340	-	0.389	-	-	-
Italy	-	-	-	-	-	0.104
Peru	0.085	-	0.092	-	-	-
Portugal	-	0.706	0.001	0.706	-	-
Sri Lanka	0.281	-	0.351	-	-	-
Sweden	0.035	-	0.090	-	-	-
Switzerland	-	-	0.004	-	-	-
UK	-	0.062	-	0.062	-	-

Notes: The table presents the composition of the synthetic control units in various specifications. "Baseline": includes all available countries in the donor pool. "No Finland": drops Finland from the donor pool. "WWI": only includes countries that lost at least 0.5% of their population in World War I (Austria, Finland, France, Germany, Italy, the United Kingdom, Australia, New Zealand, and Canada) in the donor pool.

TABLE A5

SYNTHETIC CONTROL, BALANCING, LEVELS

Unit	Population (thousands)	WWI Losses (thousands)	WWI Losses (synthetic, %)	WWI Losses (% of population)
Actual	156,192	1,700	1.09	1.09
Synthetic	106,013	24.4	0.20	0.02
Synthetic, DM	16,488	298.8	0.57	1.81
Synthetic, no Finland	121,380	26.0	0.02	0.02
Synthetic, no Finland, DM	16,479	298.5	0.57	1.81
Synthetic, WWI, DM	6,569	91.1	0.97	1.39

Notes: The table compares the characteristics of Russia with the characteristics of the synthetic control units in various specifications. WWI Losses (synthetic, %) are computed as the weighted average of percentage World War I losses of the units in the synthetic control unit. WWI Losses (% of population) are computed as the ratio of the synthetic World War I losses to the synthetic population. "DM" stands for "demeaned."

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